New Universality Class (?) in Non-Equilibrium Control Transitions in a Classical Spin Chain

Controlling Chaos

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Motivation

Controlling (classically) chaotic systems is a central problem in non-equilibrium dynamics with far reaching applications

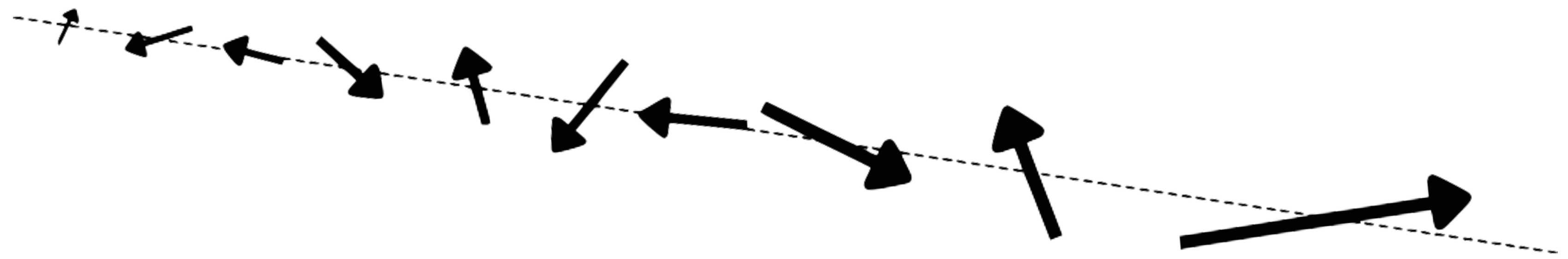
- We study behavior of a non-equilibrium phase transitions that are in the presence of an "observer" + environment
- Principals can be applied to fields such as robotics, computing, and statistical mechanics

The Model 1/3: Classical Heisenberg Spin Chain

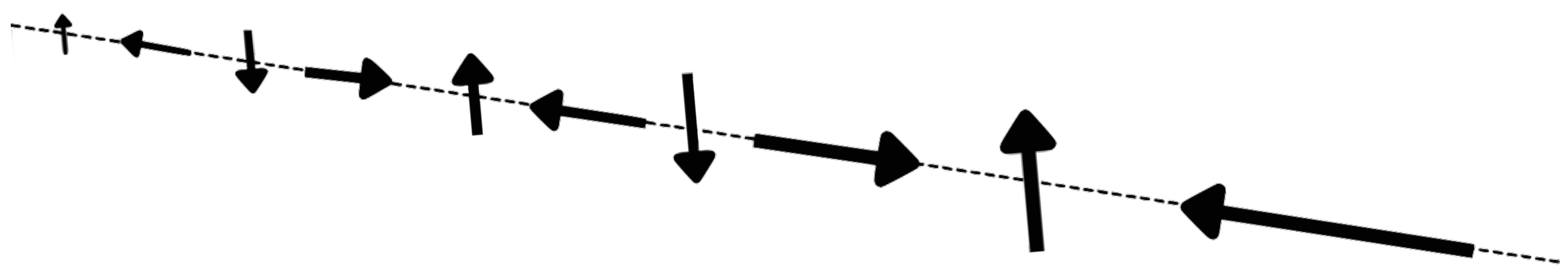
$$\boxed{H} = \sum_{i=1}^{L} \left[\left(J_x(t) S_i^x, J_y(t) S_i^y, J_z S_i^z \right) \cdot \mathbf{S}_{i+1} \right]$$

$$\frac{d\mathbf{S}_j}{dt} = \left[\mathbf{J} \circ \left(\mathbf{S}_{j-1} + \mathbf{S}_{j+1} \right) \times \mathbf{S}_j \right]$$

The dynamics of an arbitrary spin configuration are in general chaotic:



There do exist unstable fixed points such as



$$\mathbf{S}_j^0 := (0, \cos(\pi j/2), \sin(\pi j/2))$$

$$\left.\begin{aligned} \mathbf{S}_j^0(t) &= \mathbf{S}_j^0(0) \\ \mathbf{S}_j(t) &= \mathbf{S}_j^0 + \delta \mathbf{S}_j(t) \end{aligned}\right\}$$

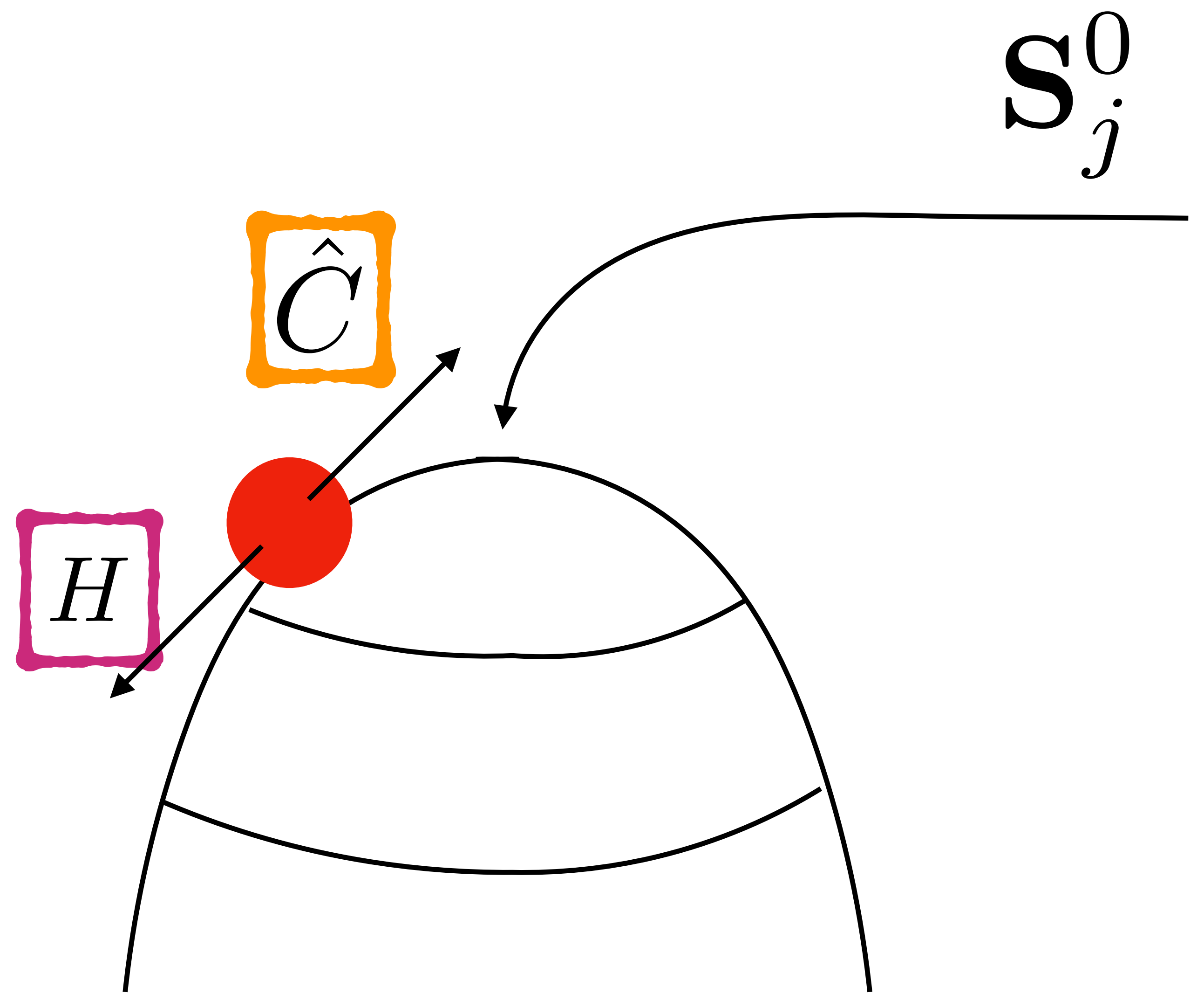
The Model 2/3: Control Map

We directly control the evolution of a spin chain by pushing it towards this unstable fixed point

Control map: $\boxed{\hat{C}}$

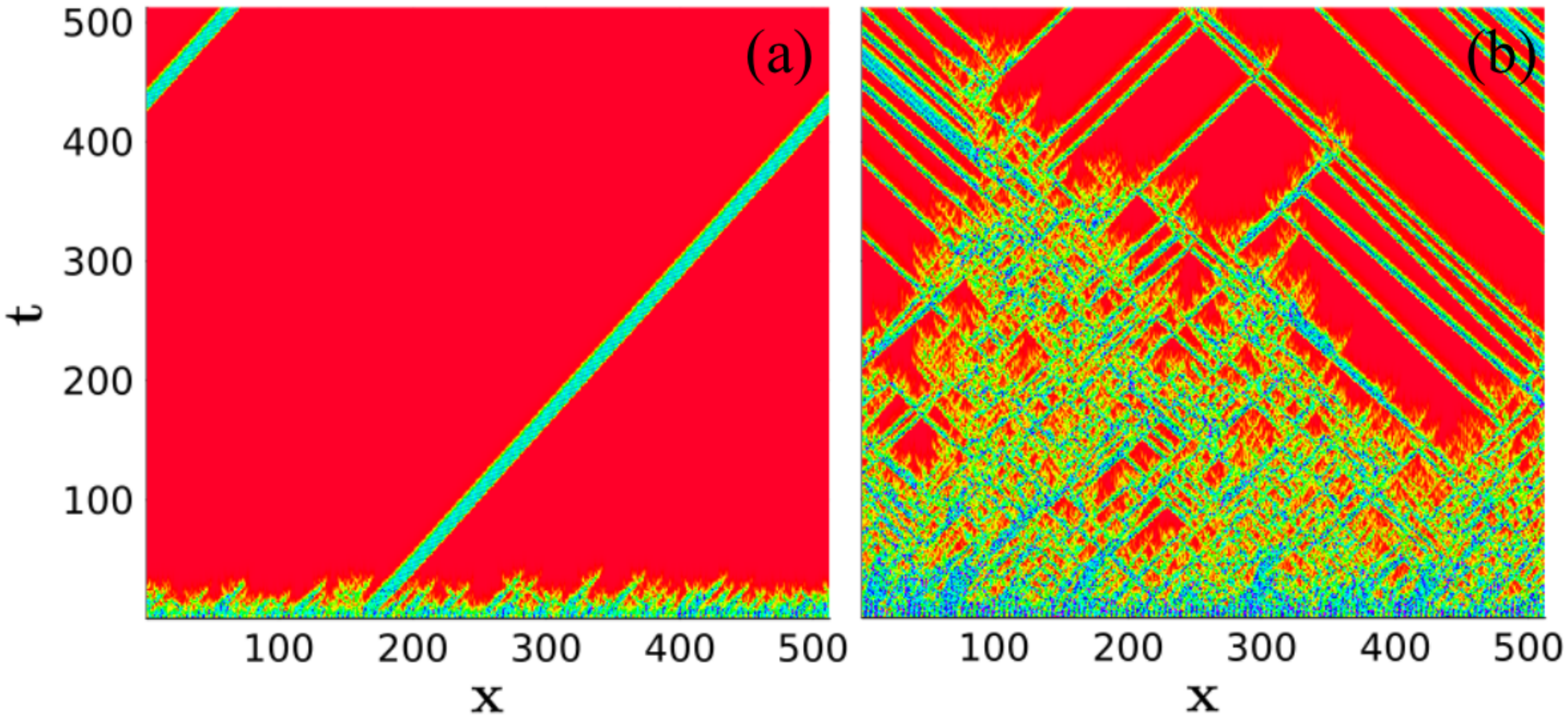
$$\hat{C}\mathbf{S}_j(t') = \frac{(1-a)\mathbf{S}_j^0 + a\mathbf{S}_j(t')}{|(1-a)\mathbf{S}_j^0 + a\mathbf{S}_j(t')|}$$

$0 < a < 1$ is our control parameter:



The Emergence of Soliton (like) Packets

$a \approx 0.6$ $\qquad$ $a = 0.7615$



$$S_{\text{diff}}(t) := \frac{1}{L}\sum_j |\delta \mathbf{S}_j(t)|$$

$$\mathbf{S}_j(t) = \mathbf{S}_j^0 + \delta \mathbf{S}_j(t)$$

Makes it very difficult do define, numerically, the critical point, or to calculate critical exponents

These "solitons" are the subject of future research

The Model 3/3: Adding Time Randomness

Random sign:

$$\begin{cases} J_x \in \{-J, J\} \\ J_y \in \{-J, J\} \\ J_z = J \end{cases}$$

And Our Order Parameter For Studying the Transition Shall be

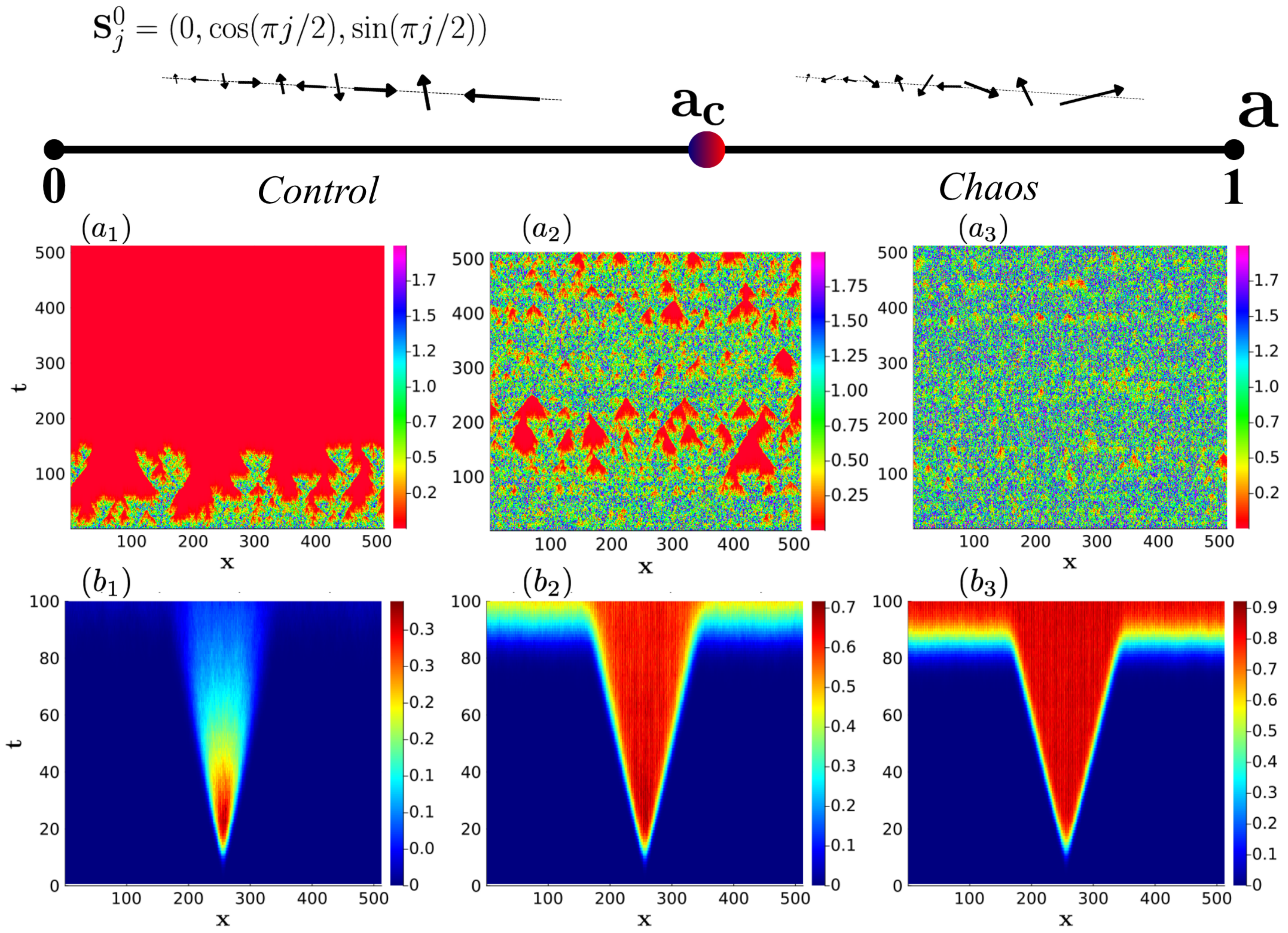
$$S_{\text{diff}}(t) := \frac{1}{L} \sum_j |\delta \mathbf{S}_j(t)|$$

$$\mathbf{S}_j(t) = \mathbf{S}_j^0 + \delta \mathbf{S}_j(t)$$

The Full Evolution Becomes:

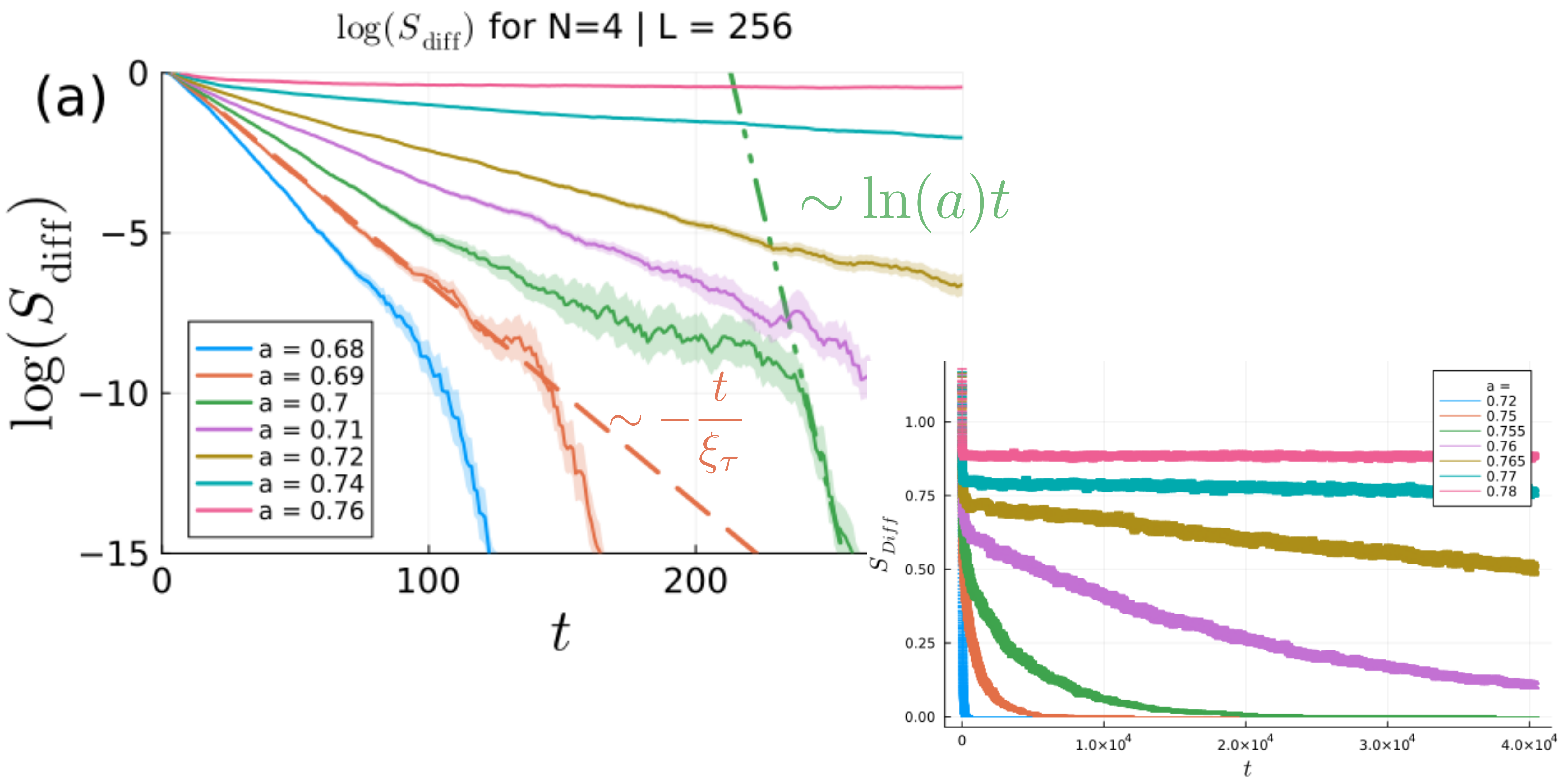
$$t : 0 \xrightarrow{\boxed{H}} \tau \xrightarrow{\boxed{\hat{C}}} \boxed{\begin{cases} J_x \in \{-J, J\} \\ J_y \in \{-J, J\} \\ J_z = J \end{cases}} \xrightarrow{\boxed{H}} 2\tau \cdots$$

The Phase Diagram

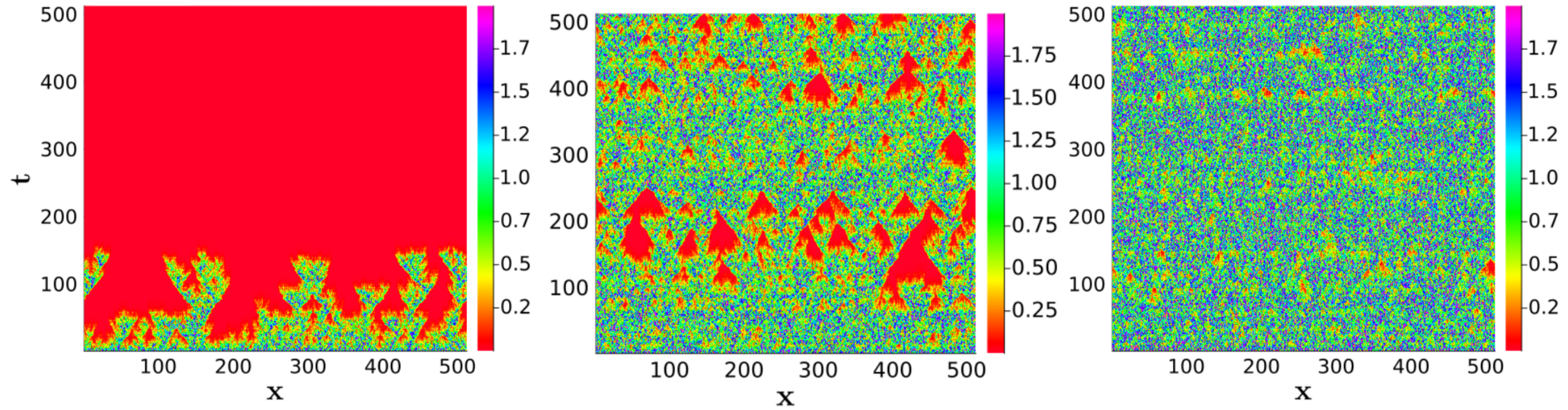


The Order Parameter: $S_{\text{diff}}(t) := \frac{1}{L}\sum_j |\delta \mathbf{S}_j(t)|$

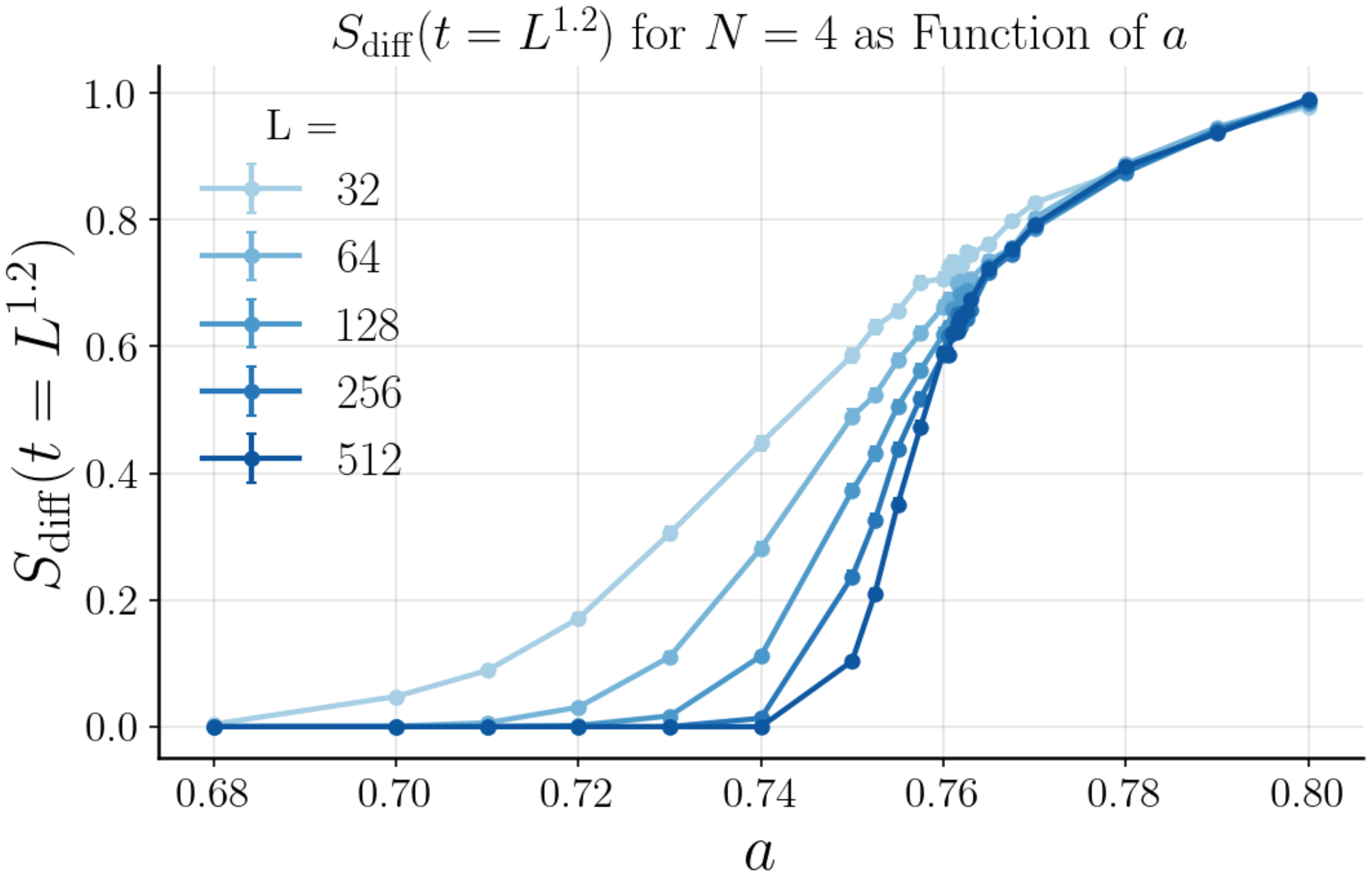
- Decays exponentially in the control phase
- Assumes and maintains a constant value in the chaotic phase



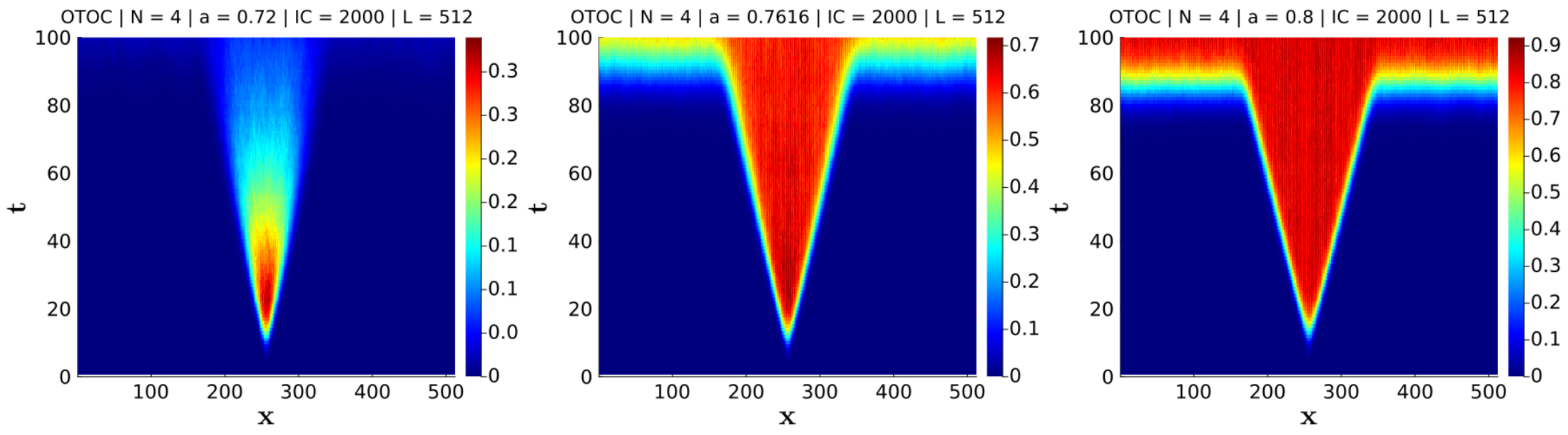
The Order Parameter Across the Transition

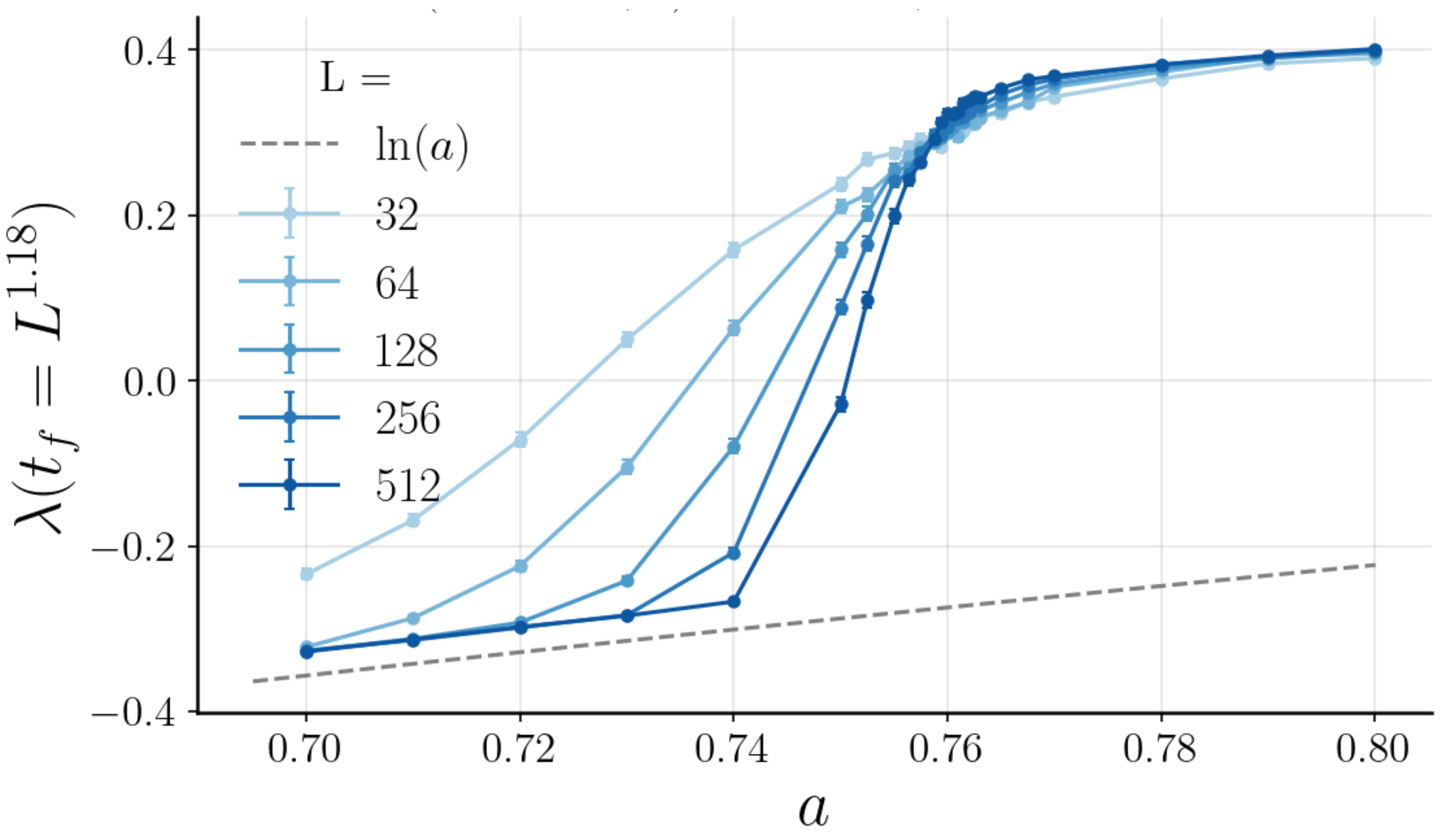


- Looks like a transition of the second order
- $a_c \approx 0.75$



Lyoponov Exponent Across the Transition





We calculate the OTOC using the method prescribed by Benettin (1)

$$\mathbf{S}_i^B(t=0) = \mathbf{S}_i^A(t=0) + \epsilon \mathbf{S}_i^r$$

$$D(x,t) := 1 - \langle \mathbf{S}_x^A \cdot \mathbf{S}_x^B \rangle \sim e^{\lambda_a t}$$

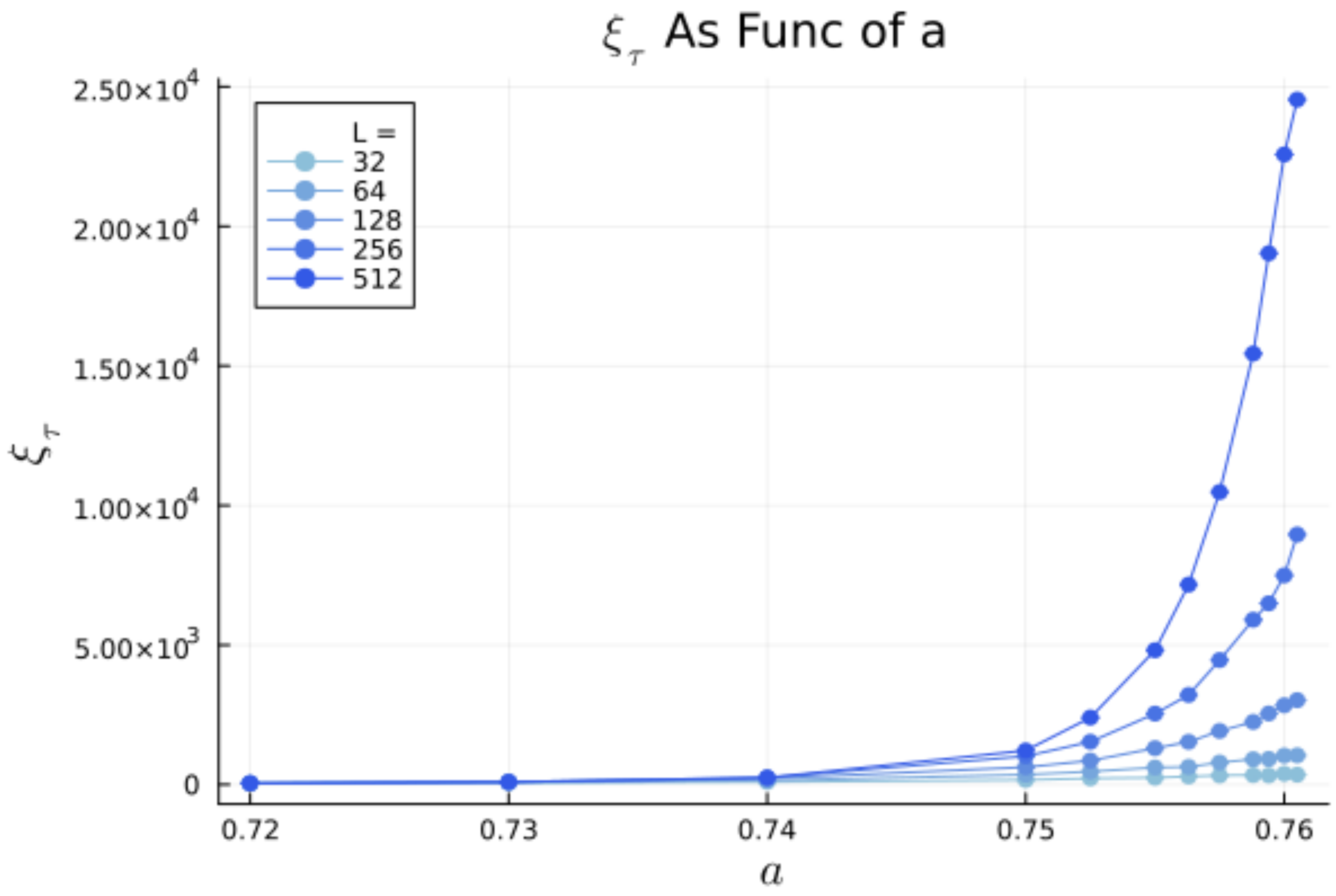
$$\lambda = \left[\frac{1}{n\tau} \sum_{j=1}^{n} \ln\left(\frac{|d_j|}{\epsilon} \right) \right] \qquad |d_j| = \sqrt{\sum_{i=1}^{L} (\mathbf{S}_i^A(t=j\tau) - \mathbf{S}_i^B(t=j\tau))^2}$$

Seems like an honest phase transition (PT), but can we be sure?

1. Giancarlo Benettin. Kolmogorov entropy and numerical experiments. Physical Review A, 14(6):2338, 1976.

The Correlation Length: Is this a Phase Transition?

$$\begin{cases} \xi_\tau \to \infty \iff \text{PT} \\ \xi_\tau \to \text{finite} \iff \text{Crossover} \end{cases} \quad \text{as } a \to a_c^-$$



This is indeed the behavior of a PT (accounting for finite size (FS) effects).

Originally it was thought that the phase transition was of the DP universality class, so we turn our attention to the matter of calculating it's critical exponents.

Behavior Near the Critical Point

We assume the following scaling behavior very near the critical point

$\Delta := (a_c - a)$

$$\boxed{S_{\text{diff},L}(t,\Delta) \sim t^{-\delta}\Phi(\Delta t^{1/\nu_\parallel}, t^{1/z})}$$

In the $t/L^z << 1$ regime this is reduced to

$$\boxed{S_{\text{diff}}(t,\Delta) \sim t^{-\beta/\nu_\parallel}\Phi(\Delta t^{1/\nu_\parallel})}$$

We can use this to independently determine

$$a_c,\ \nu_\parallel,\ \beta$$

We can then use the general behavior at

$$\Delta = 0$$

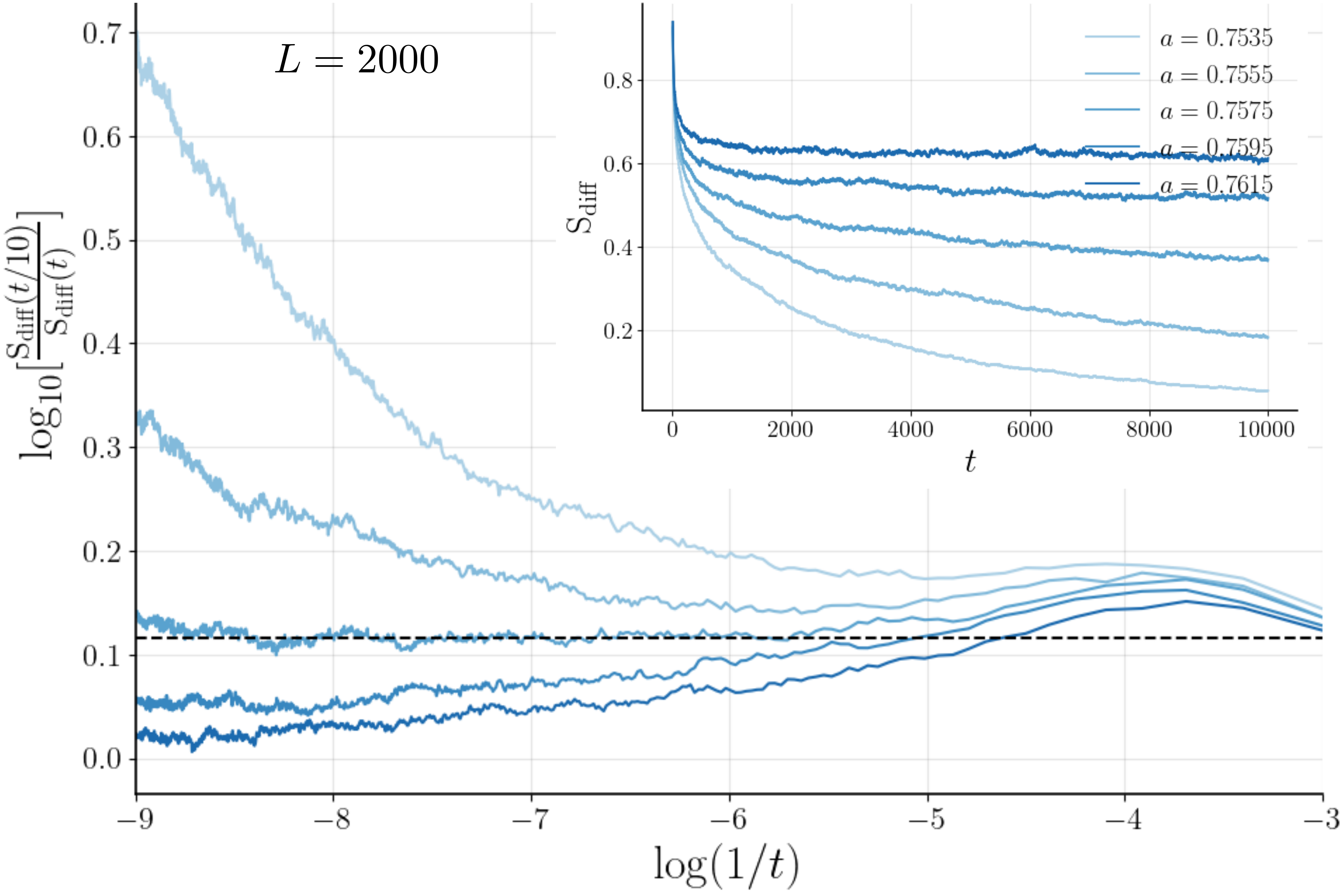
to determine

$$z$$

Note: It has been difficult to pinpoint z as the scaling technique is extremely sensitive to errors in a_c. Improving our estimate is the subject of current research.

2. Mendonça, J. R. G. (2011). Monte Carlo investigation of the critical behavior of Stavskaya's probabilistic cellular automaton. Physical Review E, 83(1), 012102. https://doi.org/10.1103/PhysRevE.83.012102

The $t/L^z << 1$ Regime: Determining a_c, $\nu_\parallel, \beta$



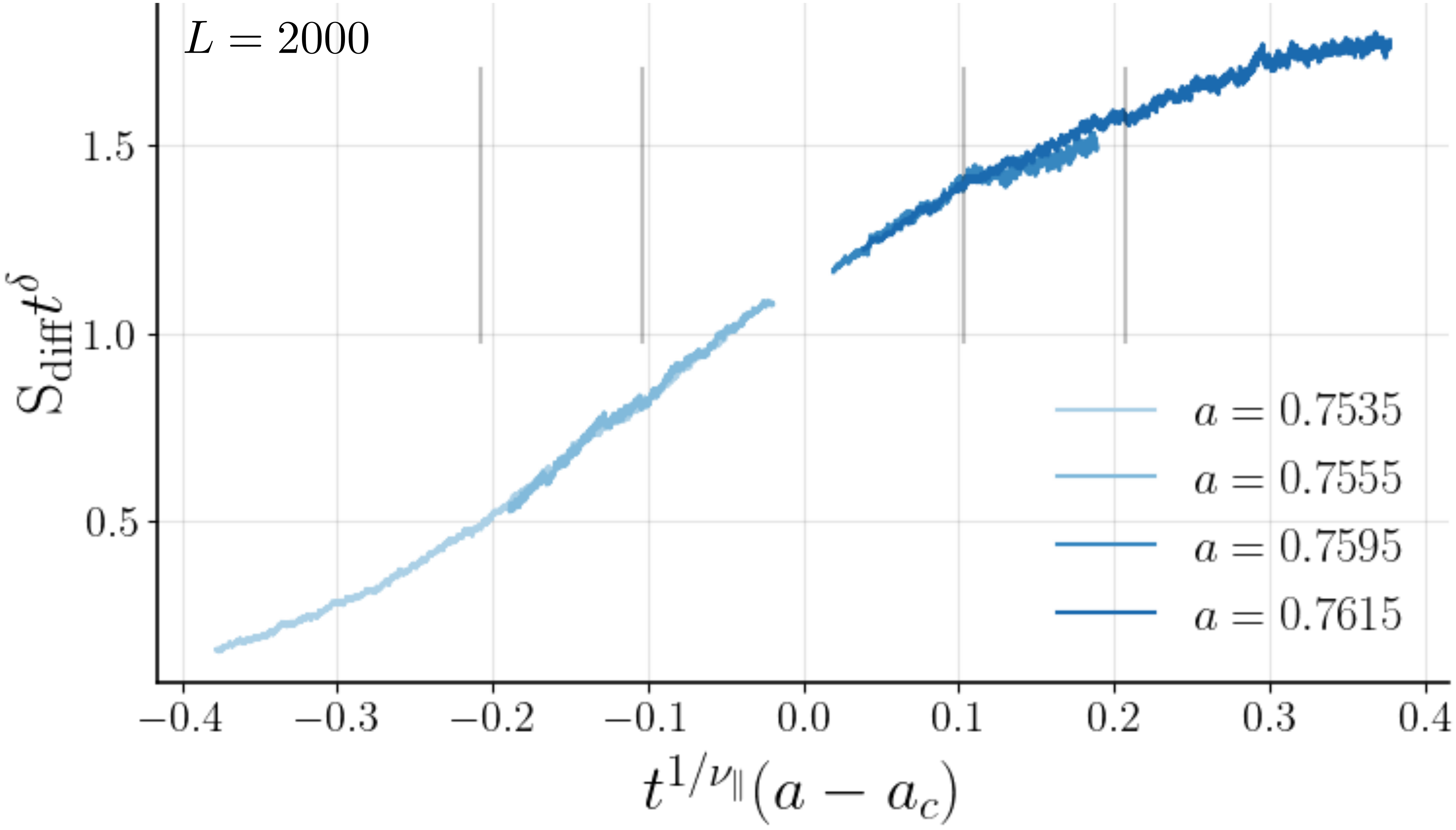
$$S_{\text{diff}}(t,\Delta) \sim t^{-\beta/\nu_\parallel}\,\Phi(\Delta t^{1/\nu_\parallel})$$

$$\boxed{\delta} := \beta/\nu_\parallel$$

$$\boxed{a_c} \approx 0.7575 \pm 0.0005$$

$$\boxed{\delta} \approx 0.116 \pm 0.005$$

The $t/L^z << 1$ Regime: Determining $a_c,\ \nu_\parallel, \beta$

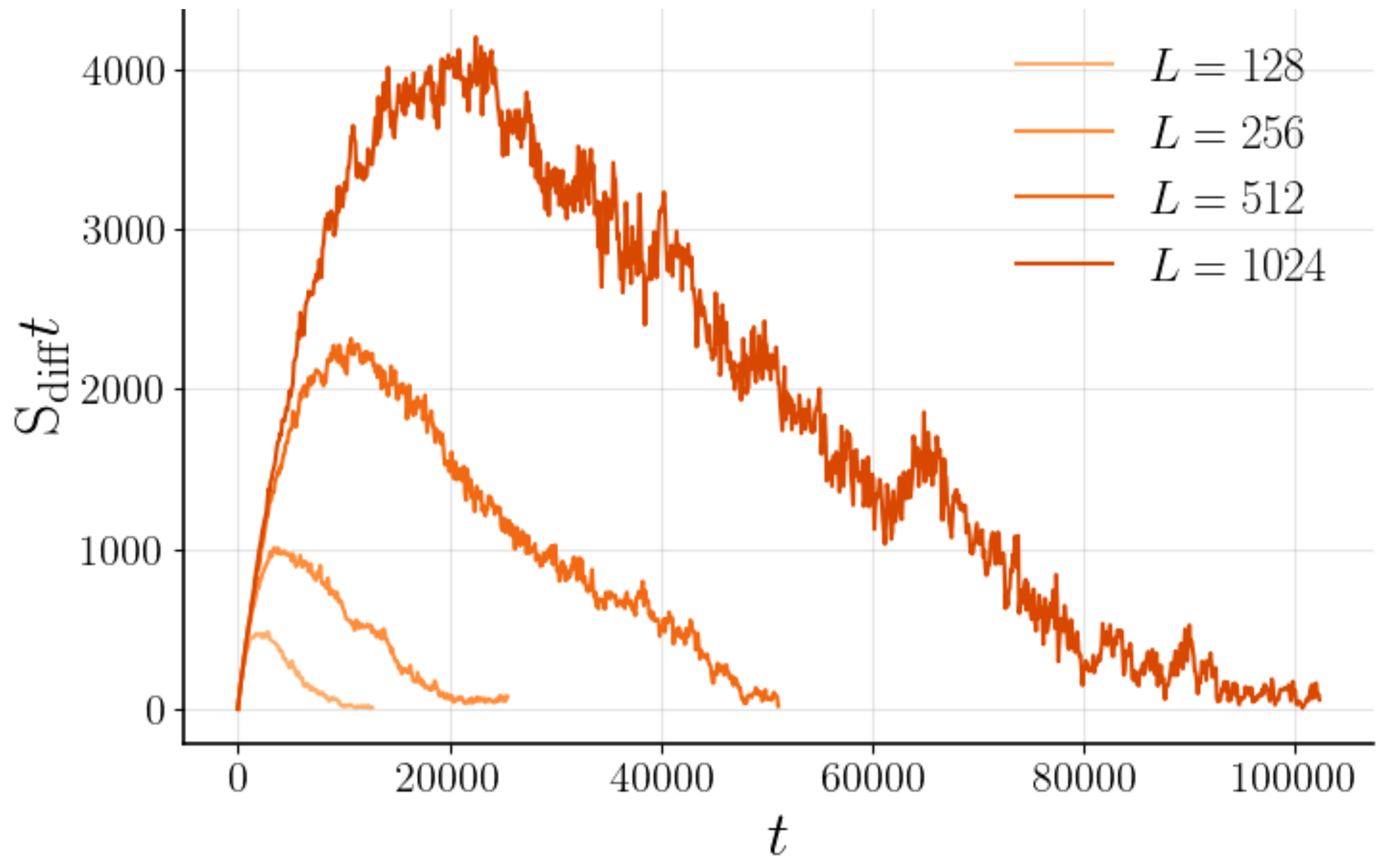


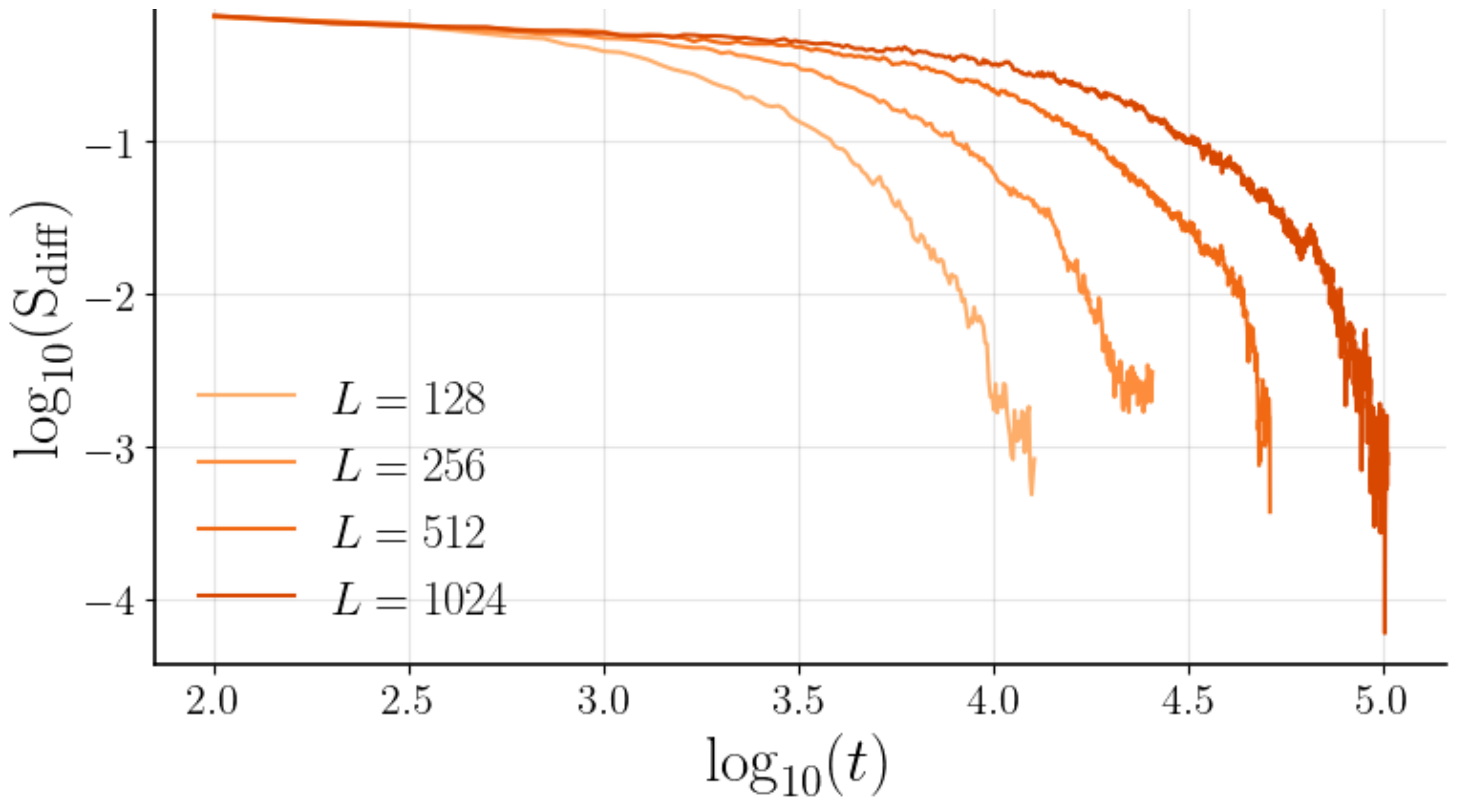
$$S_{\mathrm{diff}}(t, \Delta) \sim t^{-\beta/\nu_\parallel} \Phi(\Delta t^{1/\nu_\parallel})$$

$$\boxed{a_c} = 0.7575$$
$$\boxed{\delta} = 0.116$$

$$\boxed{\nu_\parallel} \approx 2.025 \pm 0.005$$
$$\Longrightarrow \boxed{\beta} = \nu_\parallel \delta \approx 0.23 \pm 0.01$$

Estimating z

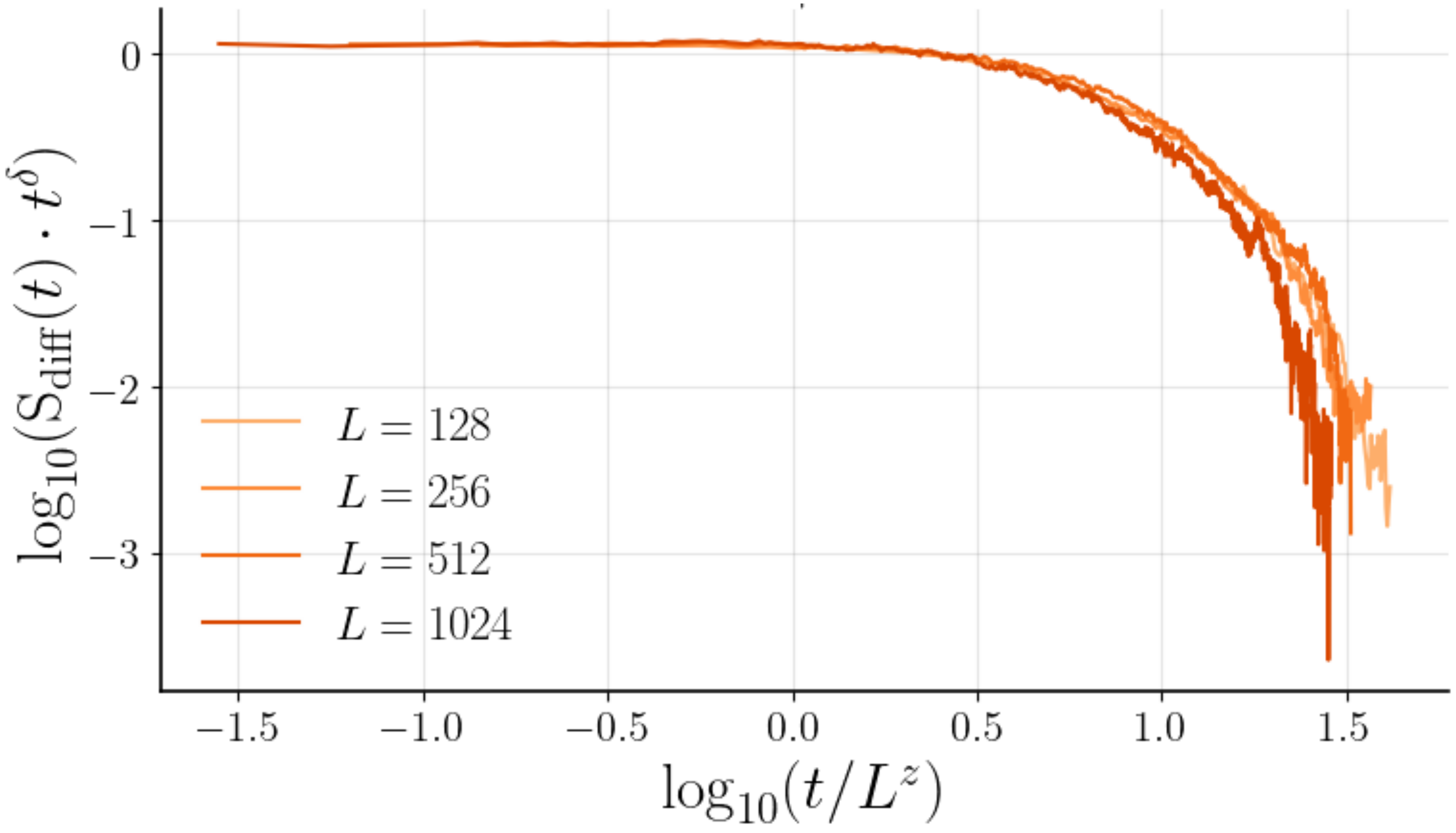




$$S_{\text{diff}} \sim t^{-\delta}\Phi(t/L^{z})$$

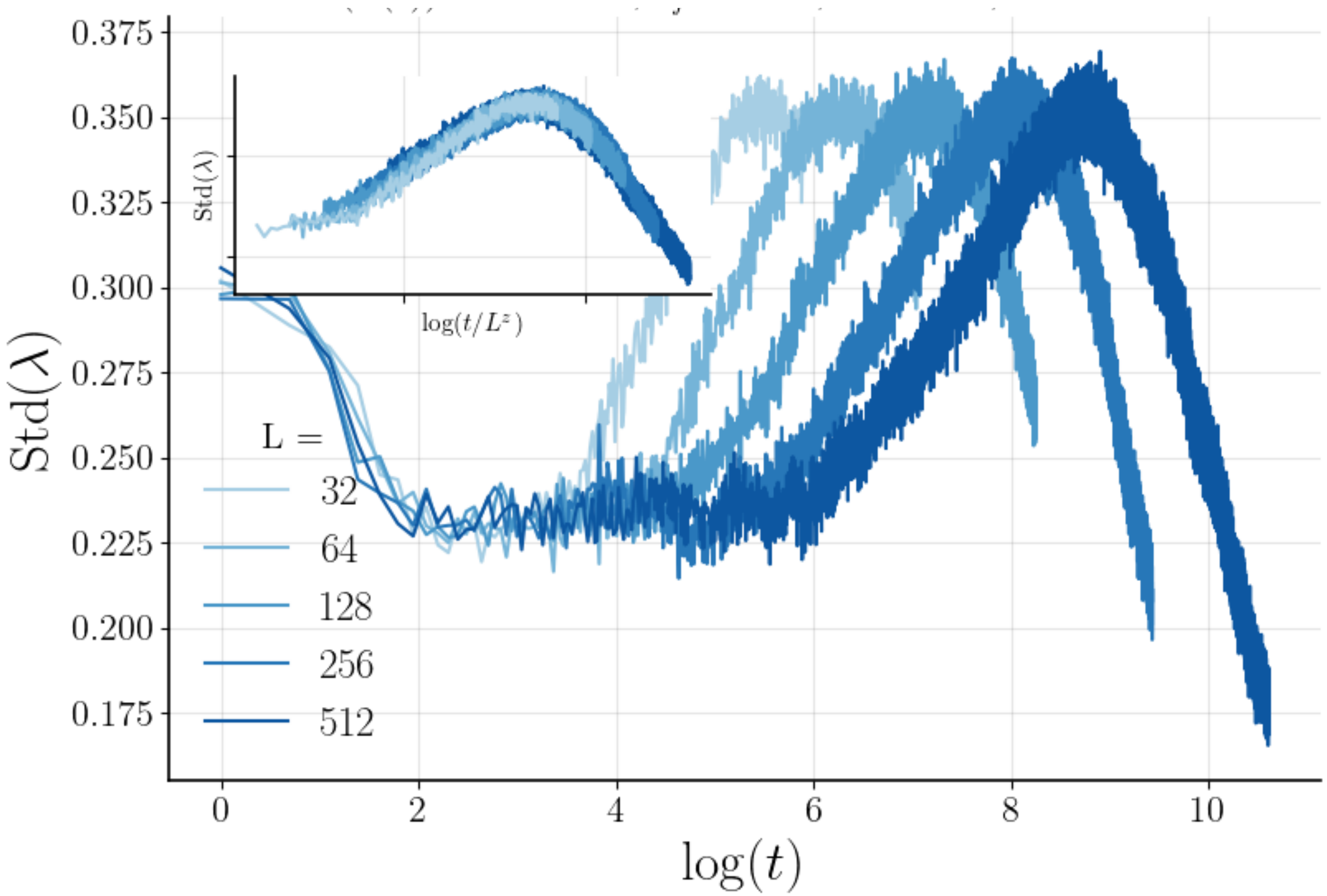
$\boxed{\delta} = 0.116$

$\boxed{\boxed{z} \approx 1.15 \pm 0.15}$



2. Mendonça, J. R. G. (2011). Monte Carlo investigation of the critical behavior of Stavskaya's probabilistic cellular automaton. Physical Review E, 83(1), 012102. https://doi.org/10.1103/PhysRevE.83.012102

Estimating z Using Lyapunov

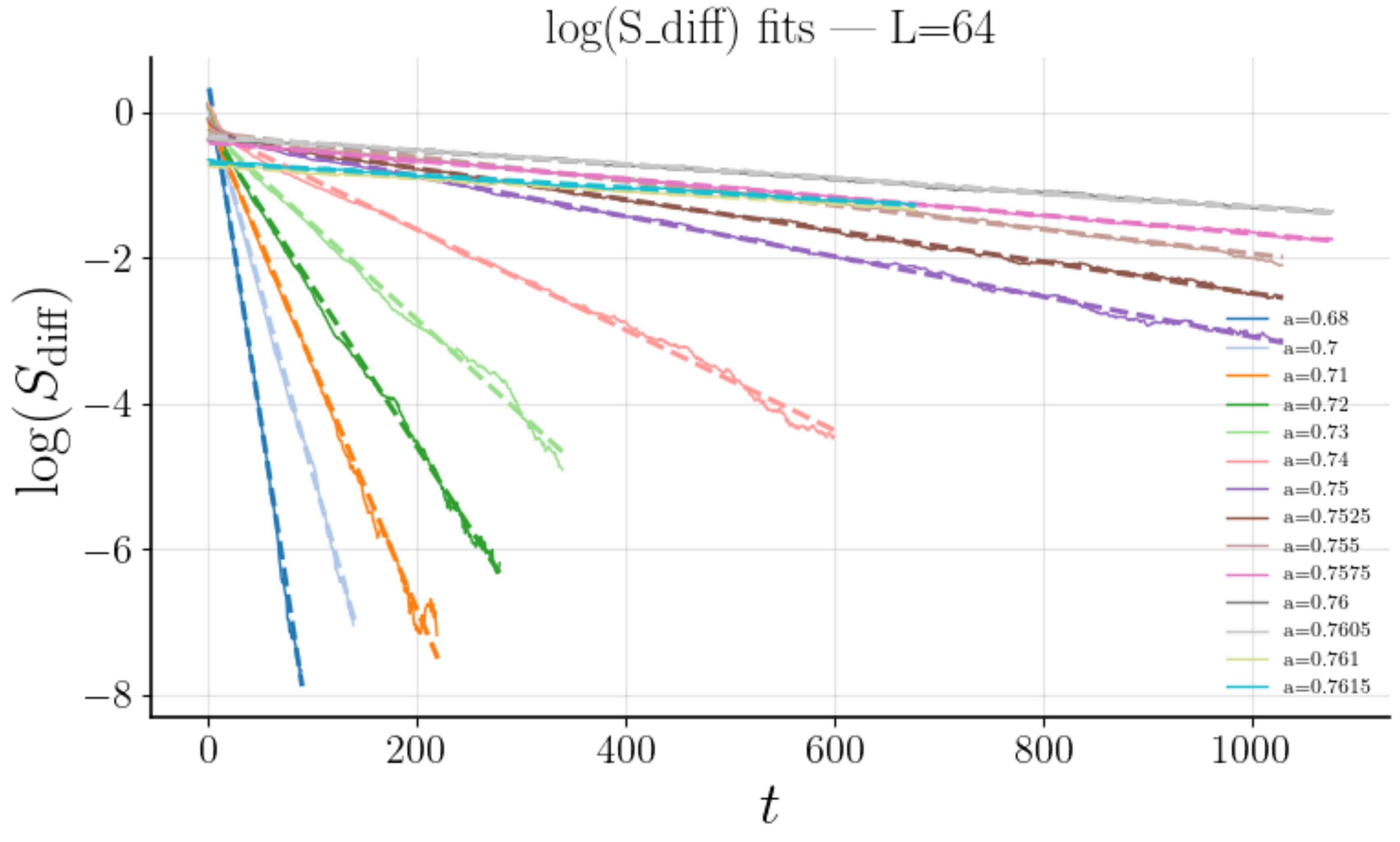


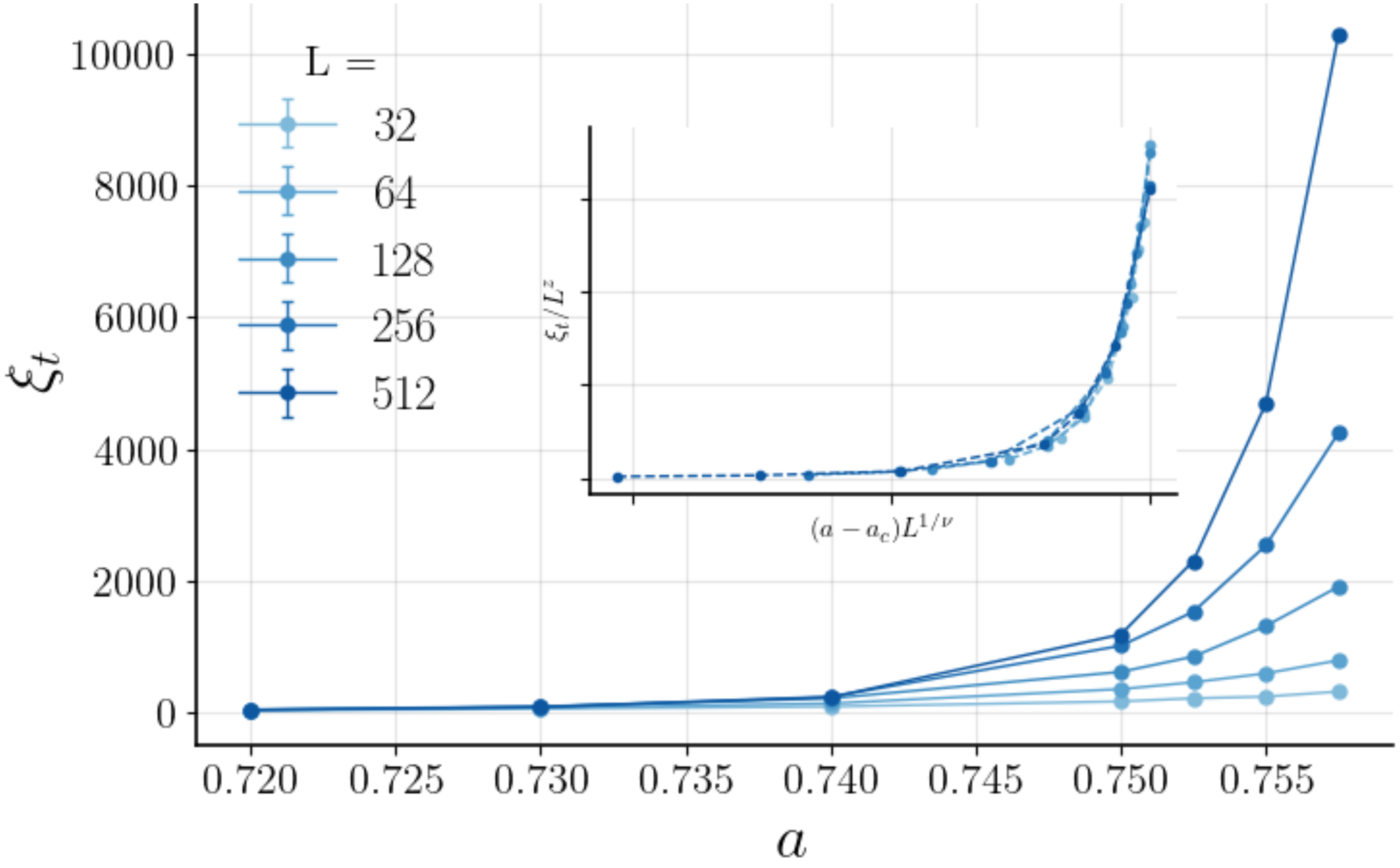
$$\mathrm{Std}(\lambda(a = a_c, t)) \sim g(t/L^z)$$

$$\boxed{a_c} = 0.7575$$

$$\boxed{z} \approx 1.18 \pm 0.12$$

Estimating z Using ξ_τ





$$\xi_\tau \sim L^z f[(a-a_c)L^{1/\nu}]$$

$$\boxed{\nu} := \nu_\parallel / z \; (= \nu_\perp)$$

$$\boxed{a_c} = 0.7575$$

$$\boxed{z} \approx 1.3 \pm 0.1$$

Results of Near Critical Point Analysis

The order parameter

$$S_{\text{diff},L}(t,\Delta) \sim t^{-\delta}\Phi(\Delta t^{1/\nu_\parallel}, t^{1/z})$$

$\boxed{a_c} \approx 0.7575 \pm 0.0005$

$\boxed{\nu_\parallel} \approx 2.025 \pm 0.005$

$\boxed{\beta} \approx 0.23 \pm 0.01$

$\boxed{z} \approx 1.15 \pm 0.15$

The STD of the Lyapunov exponent

$$\text{Std}(\lambda(a = a_c, t)) \sim g(t/L^z)$$

$\boxed{z} \approx 1.18 \pm 0.12$

Temporal correlation length

$$\xi_\tau \sim L^z f[(a - a_c)L^{1/\nu}]$$

$\boxed{z} \approx 1.3 \pm 0.1$

$\boxed{\nu} := \nu_\parallel / z \approx 1.6 \pm 0.2$

Values for DP are: $\nu_\parallel \approx 1.734$, $z \approx 1.58$, $\beta \approx 0.28$

New Universality Class? Time Random DP?

We wish to consider the possibility that we have stumbled upon a new Universality Class

Random sign:

$$\begin{cases} J_x \in \{-J, J\} \\ J_y \in \{-J, J\} \\ J_z = J \end{cases}$$

From the Renormalization Group Theory, one can reasonably expect this operation to alter the universality class of the system

To test this, we consider a toy model which is known to be of the DP universality class: The Stavskaya Probabilistic Cellular Automaton

The Comparison Model: Stavskaya Probabilistic Cellular Automaton

The system: $\eta(t) = (\eta_1(t), \ldots, \eta_L(t)) \in \{0,1\}^L$

With fixed probability $\epsilon \in [0,1], \; \eta_i(t+1) = 1$ (1)

Otherwise $\eta_i(t+1) = \eta_{i-1}(t) \cdot \eta_i(t)$ (2)

The control parameter: ϵ

The order parameter:

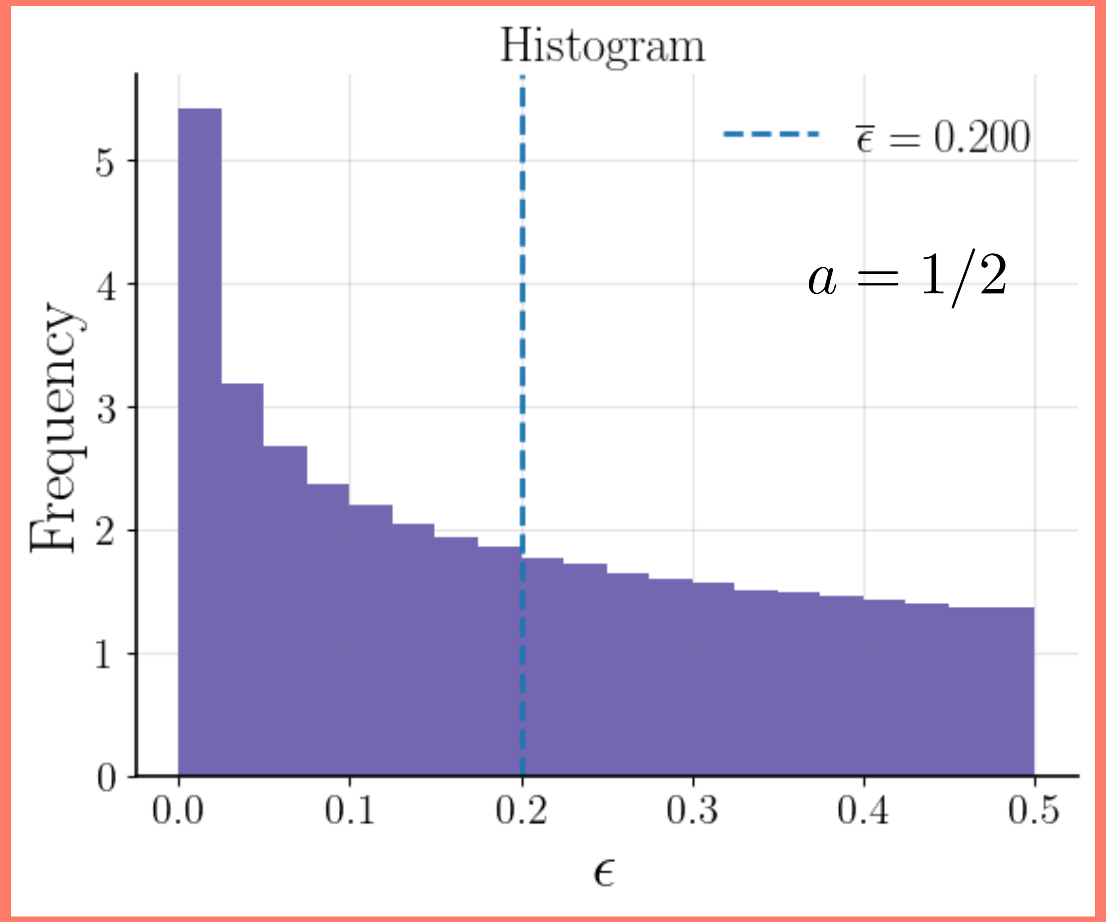
$$1 - \rho = 1 - \frac{1}{L}\sum_{i=1}^{L} \eta_i(t)$$

"Broad" Distribution

$\epsilon(t) = a(X)^n$

$X = \mathrm{Unif}(0,1)$

$$\bar{\epsilon} = \frac{a}{n+1}$$



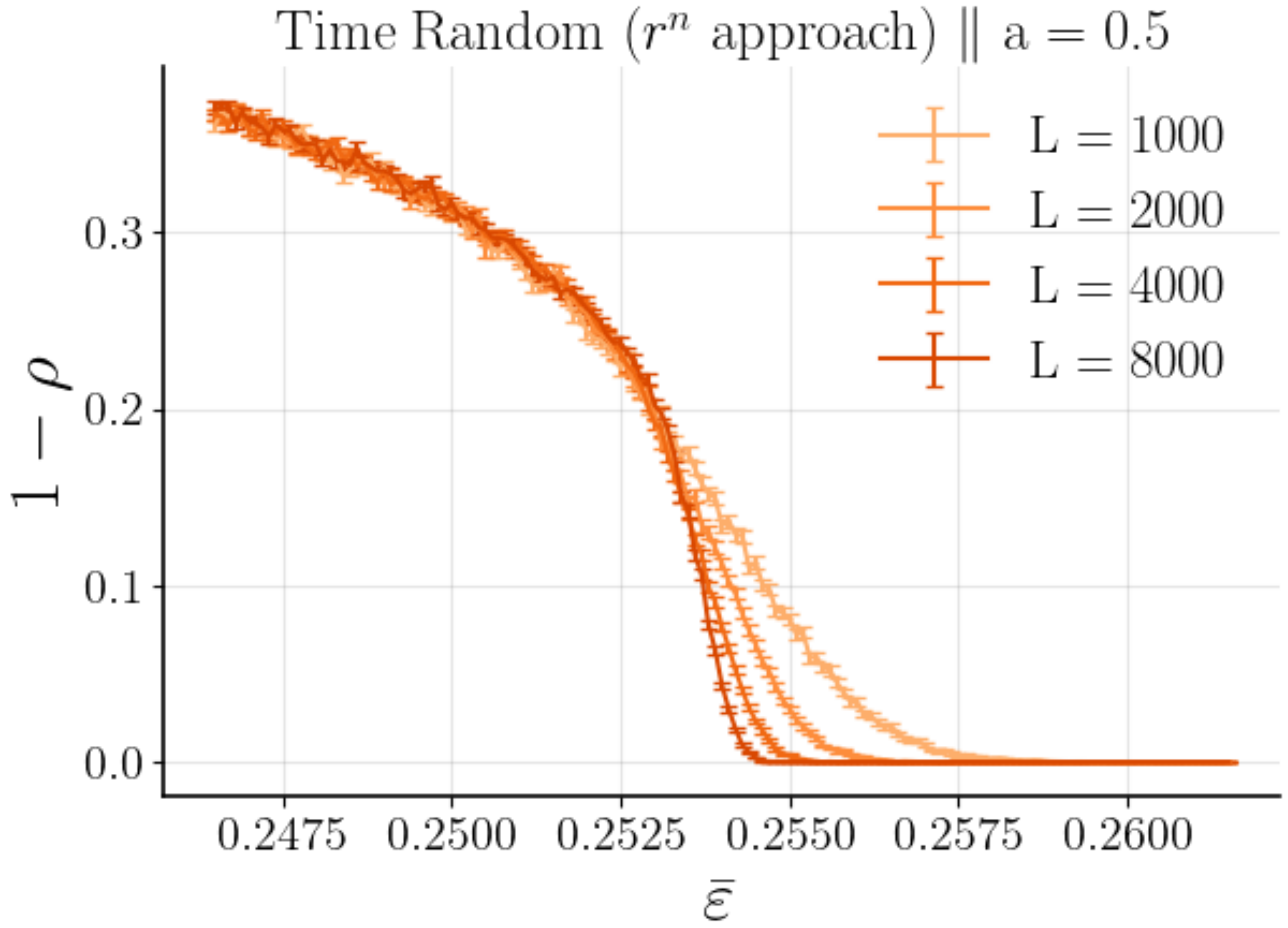
Binary Distribution

$$\epsilon = 2\bar{\epsilon}\{0,1\}$$

cricket cricket

2. Mendonça, J. R. G. (2011). Monte Carlo investigation of the critical behavior of Stavskaya's probabilistic cellular automaton. Physical Review E, 83(1), 012102. https://doi.org/10.1103/PhysRevE.83.012102

The Broad Distribution



$$1-\rho(t;\Delta) \sim t^{-\beta/\nu_\parallel} f(\Delta t^{1/\nu_\parallel}, t^{\nu_\perp/\nu_\parallel})$$

$$\delta := \beta/\nu_\parallel$$

$$(1-\rho(t;\Delta))t^\delta \sim f(\Delta t^{1/\nu_\parallel}, t^{\nu_\perp/\nu_\parallel})$$

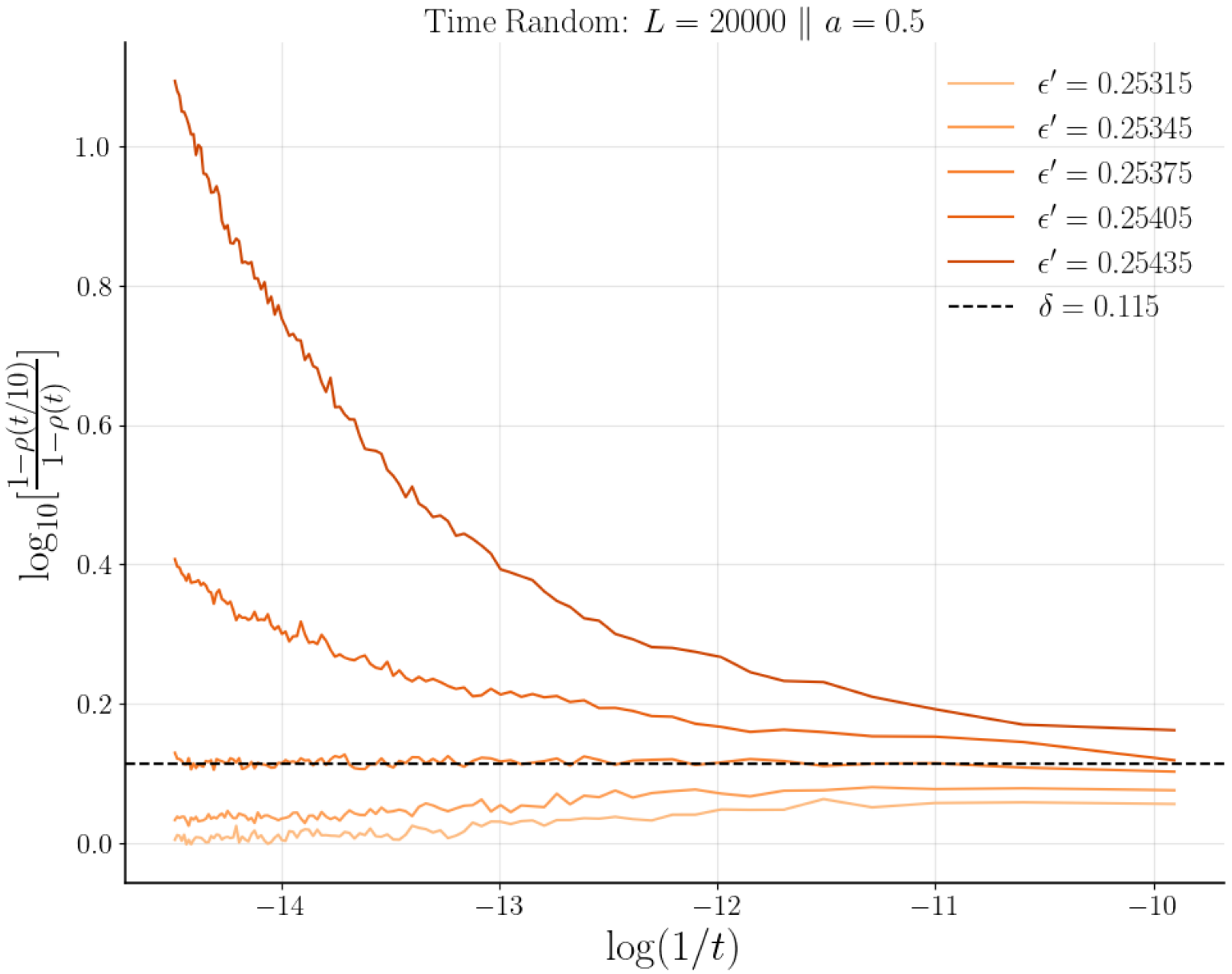
The Broad Distribution

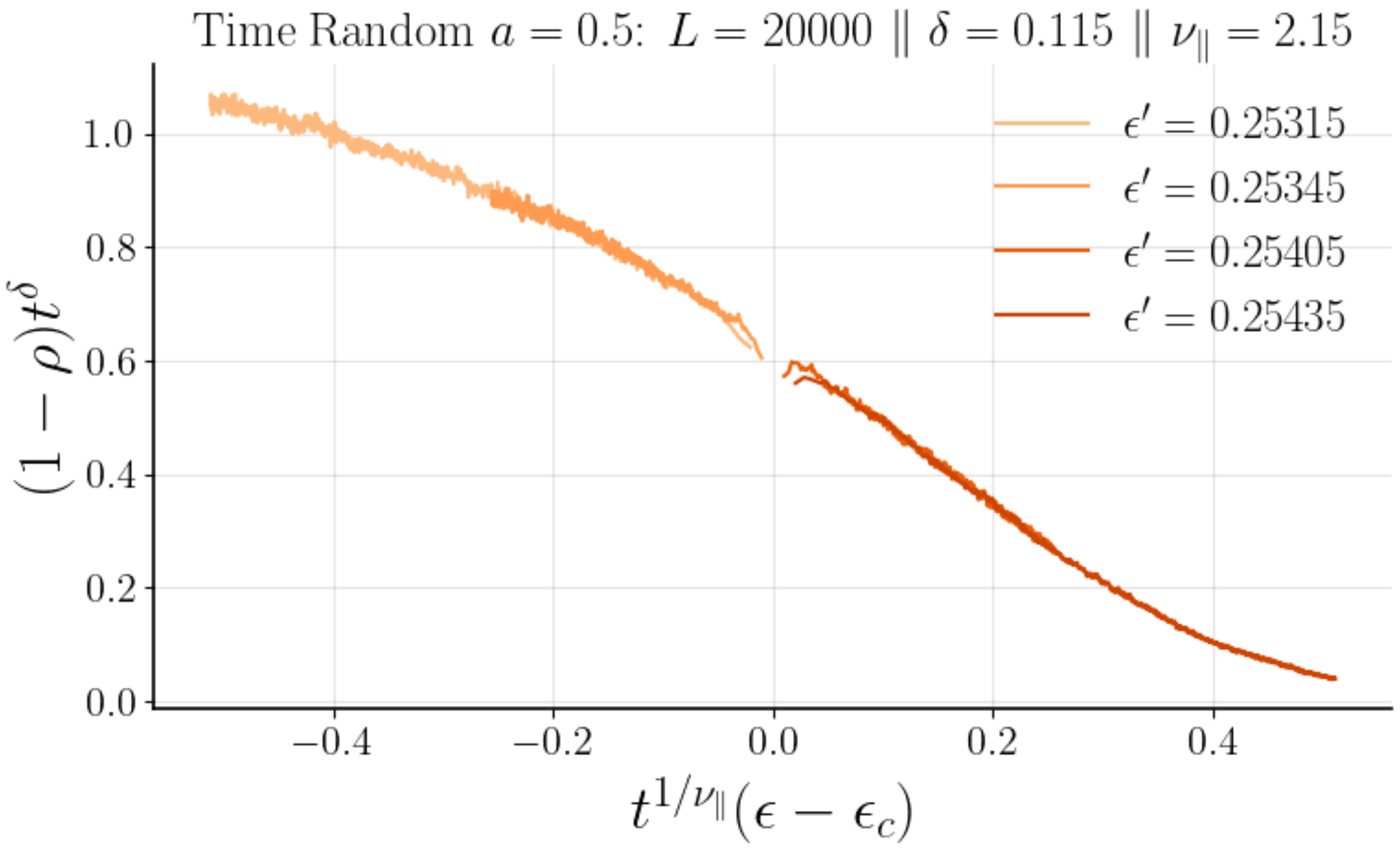
$\boxed{a_c} \approx 0.252375$

$\boxed{\nu_\parallel} \approx 2.15 \pm 0.05$

$\boxed{\delta} \approx 0.115 \pm 0.01$

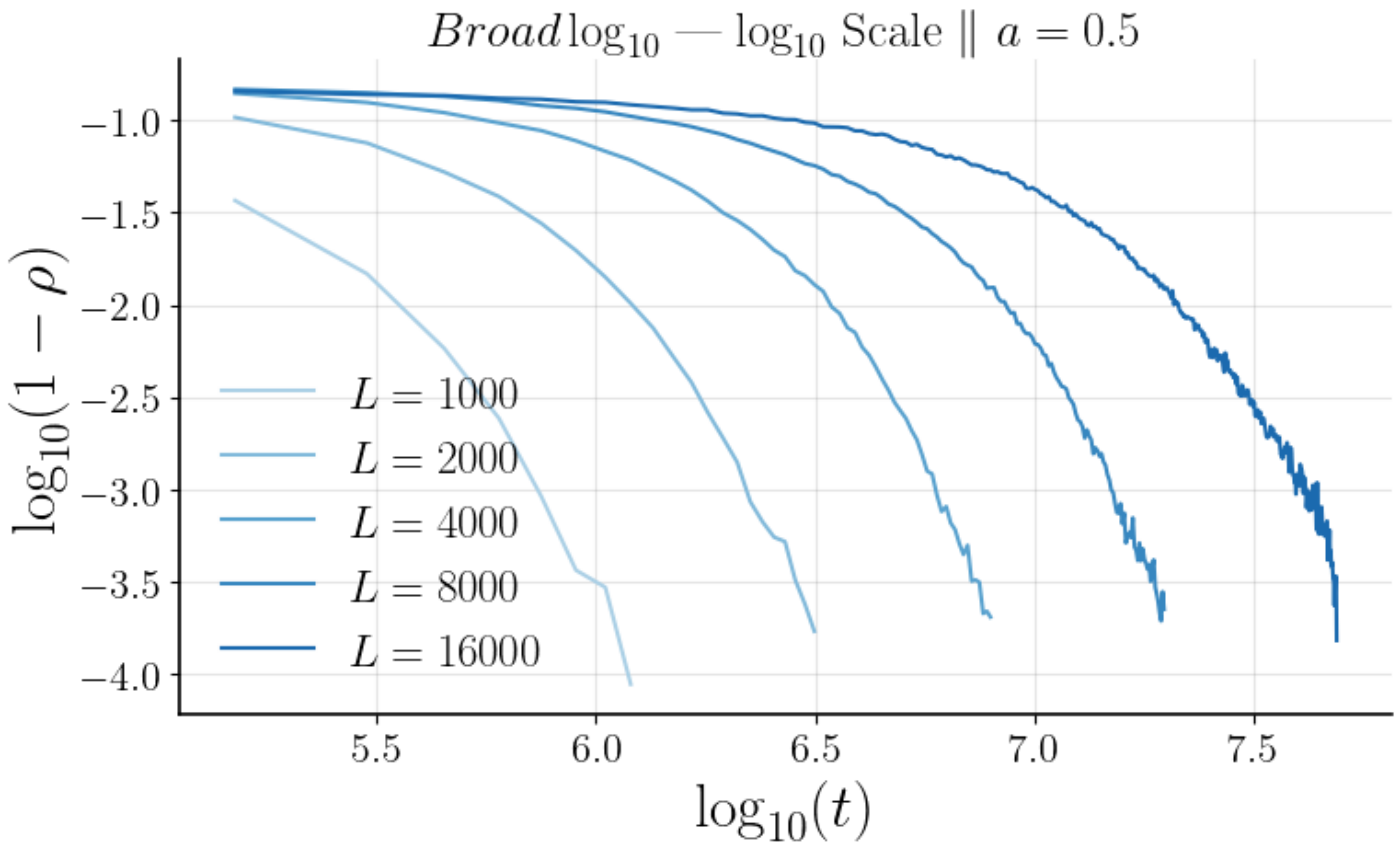
$\boxed{\beta} \approx 0.25 \pm 0.02$

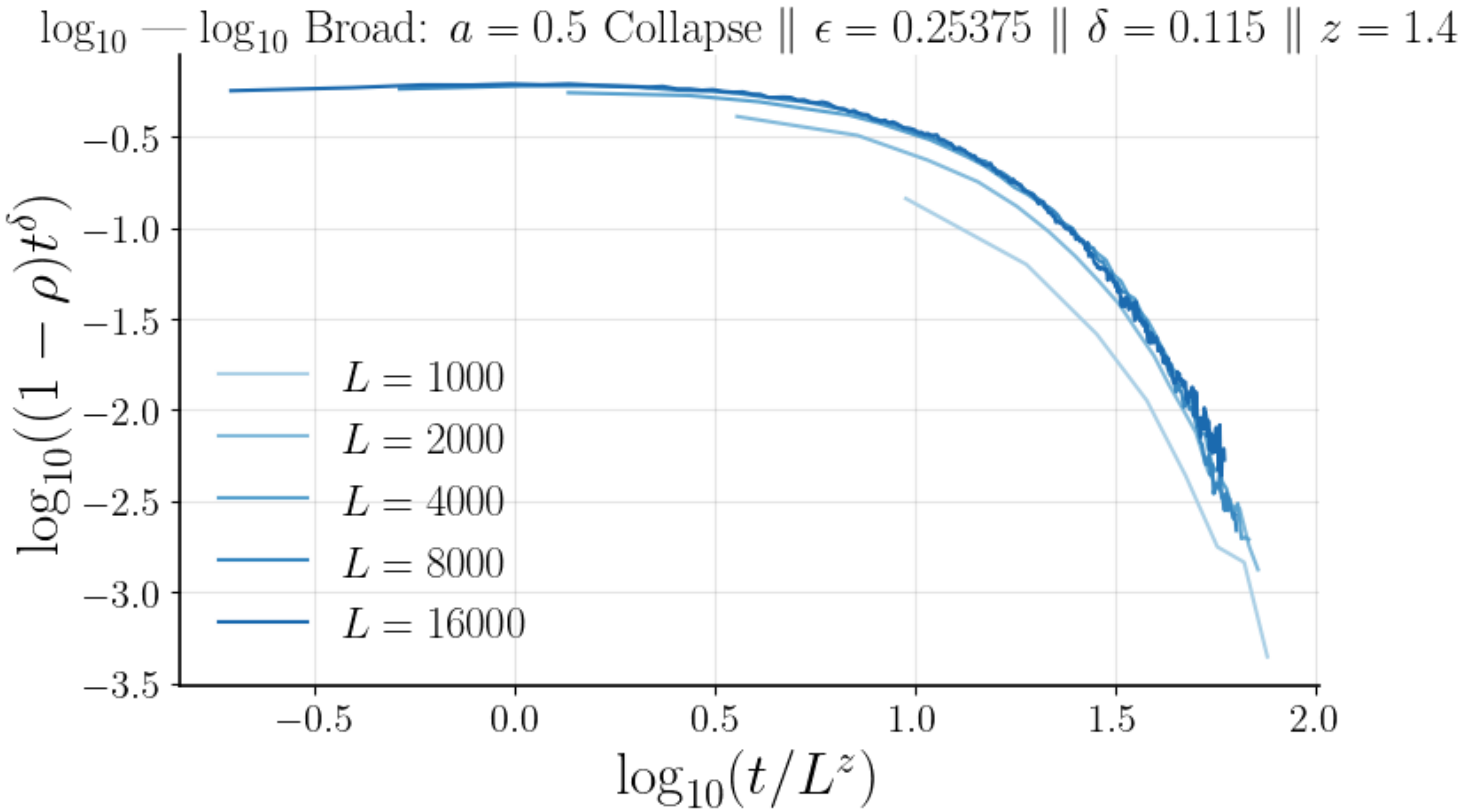




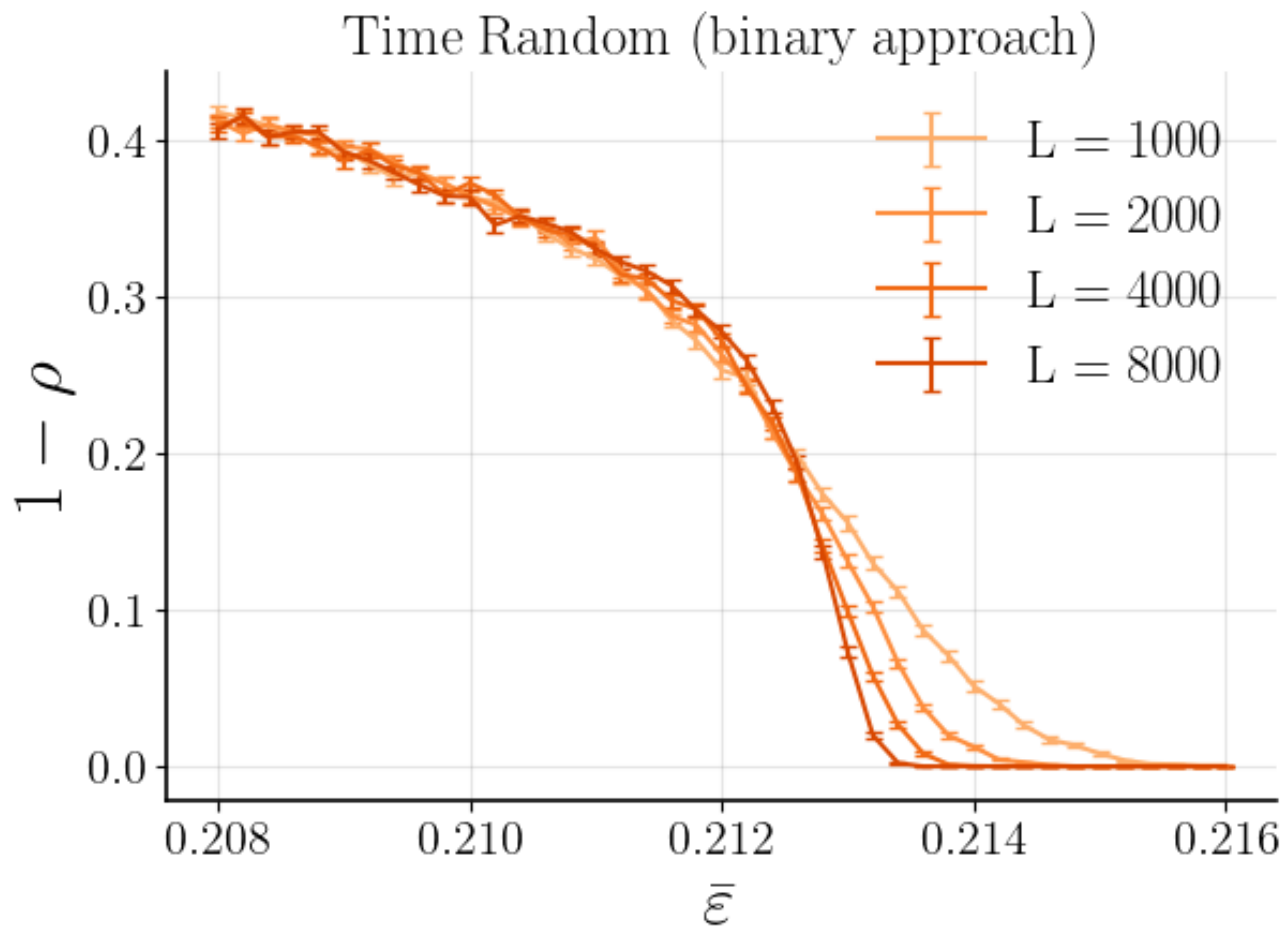
The Broad Distribution

$\boxed{\delta} = 0.115 \quad \boxed{z} = 1.4$





Binary Distribution

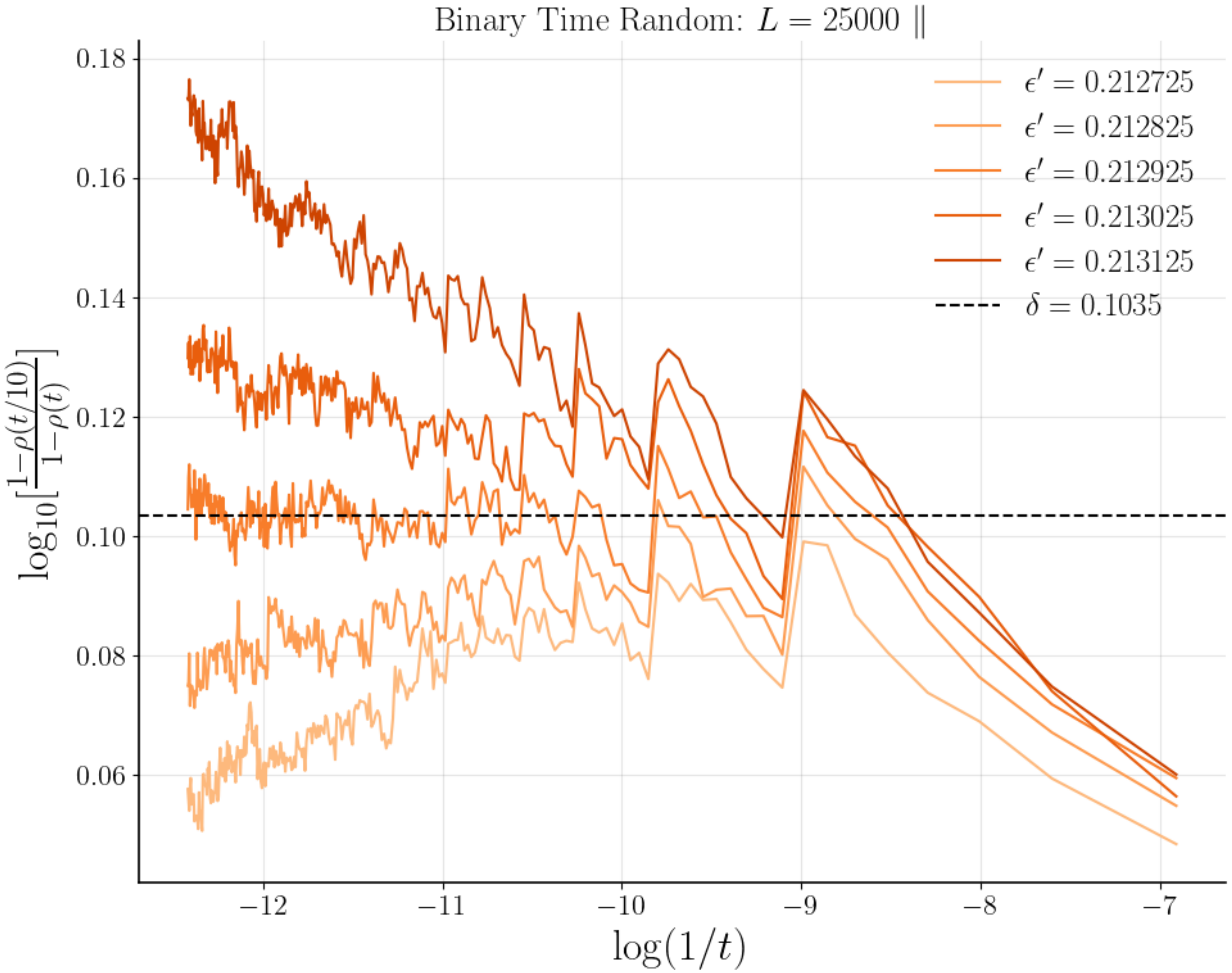


$$1-\rho(t;\Delta) \sim t^{-\beta/\nu_\parallel} f(\Delta t^{1/\nu_\parallel}, t^{\nu_\perp/\nu_\parallel})$$

$$\delta := \beta/\nu_\parallel$$

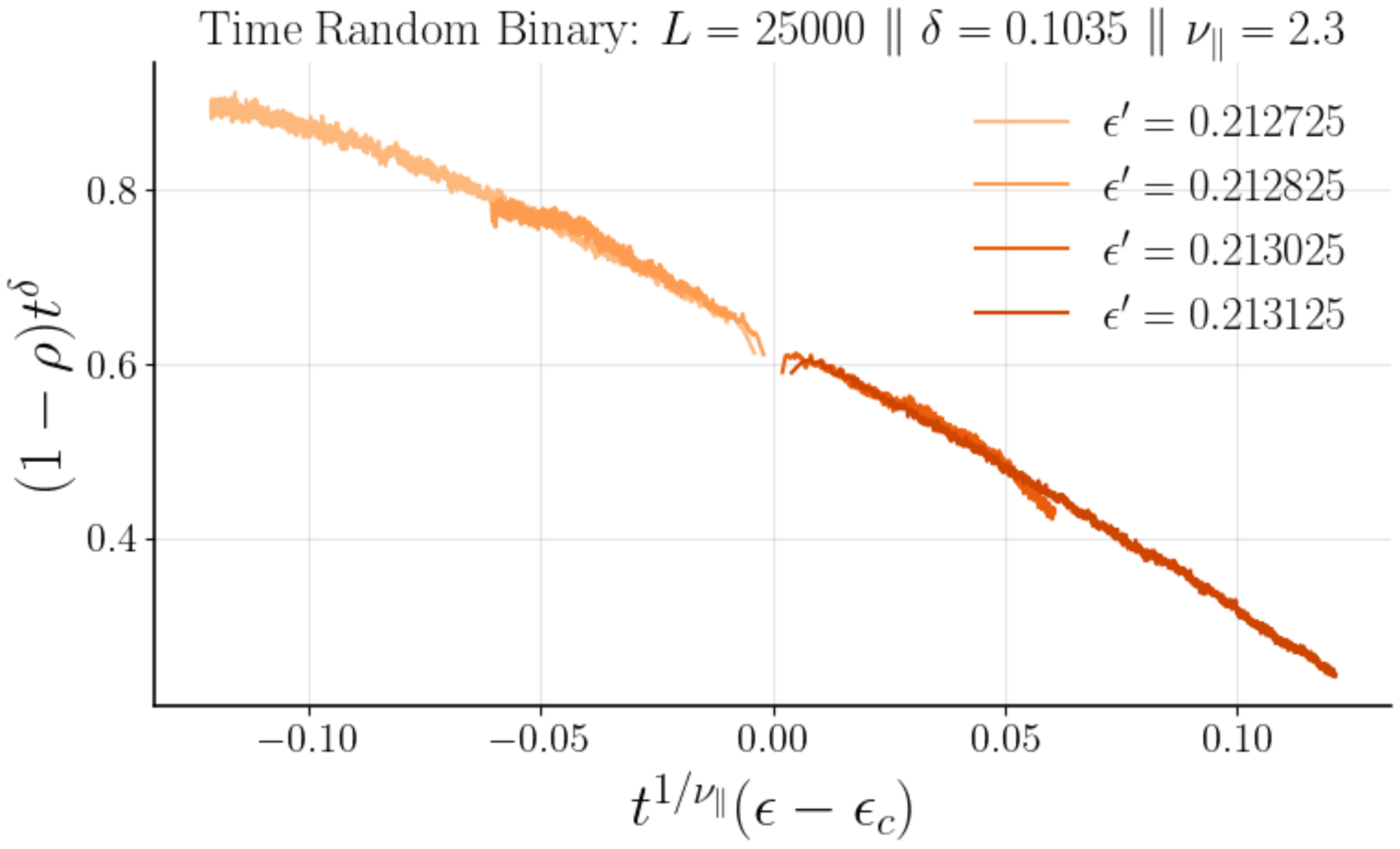
$$(1-\rho(t;\Delta))t^{\delta} \sim f(\Delta t^{1/\nu_\parallel}, t^{\nu_\perp/\nu_\parallel})$$

The Binary Distribution



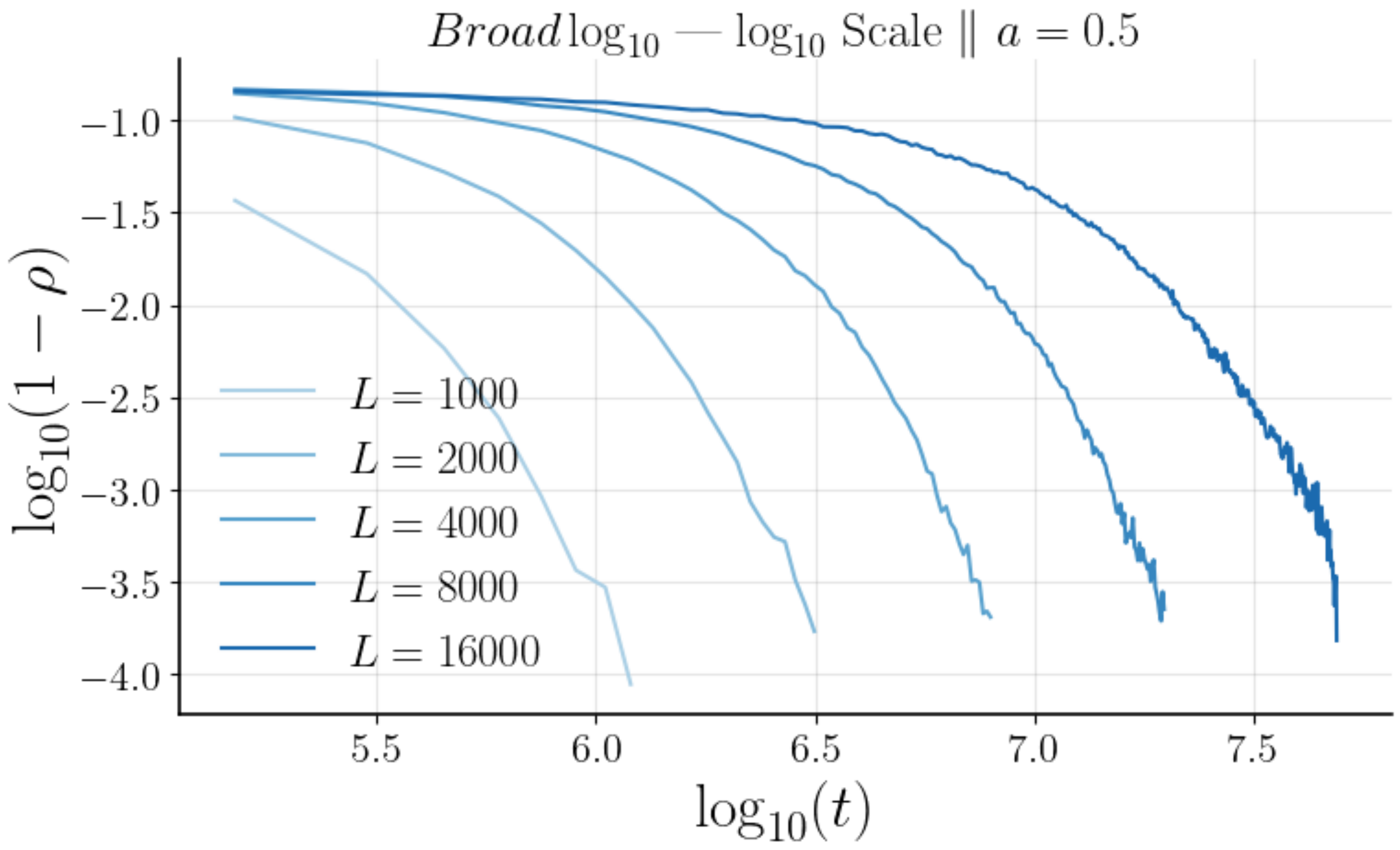
$\boxed{a_c} \approx 0.212925$ $\boxed{\delta} \approx 0.1035 \pm 0.002$

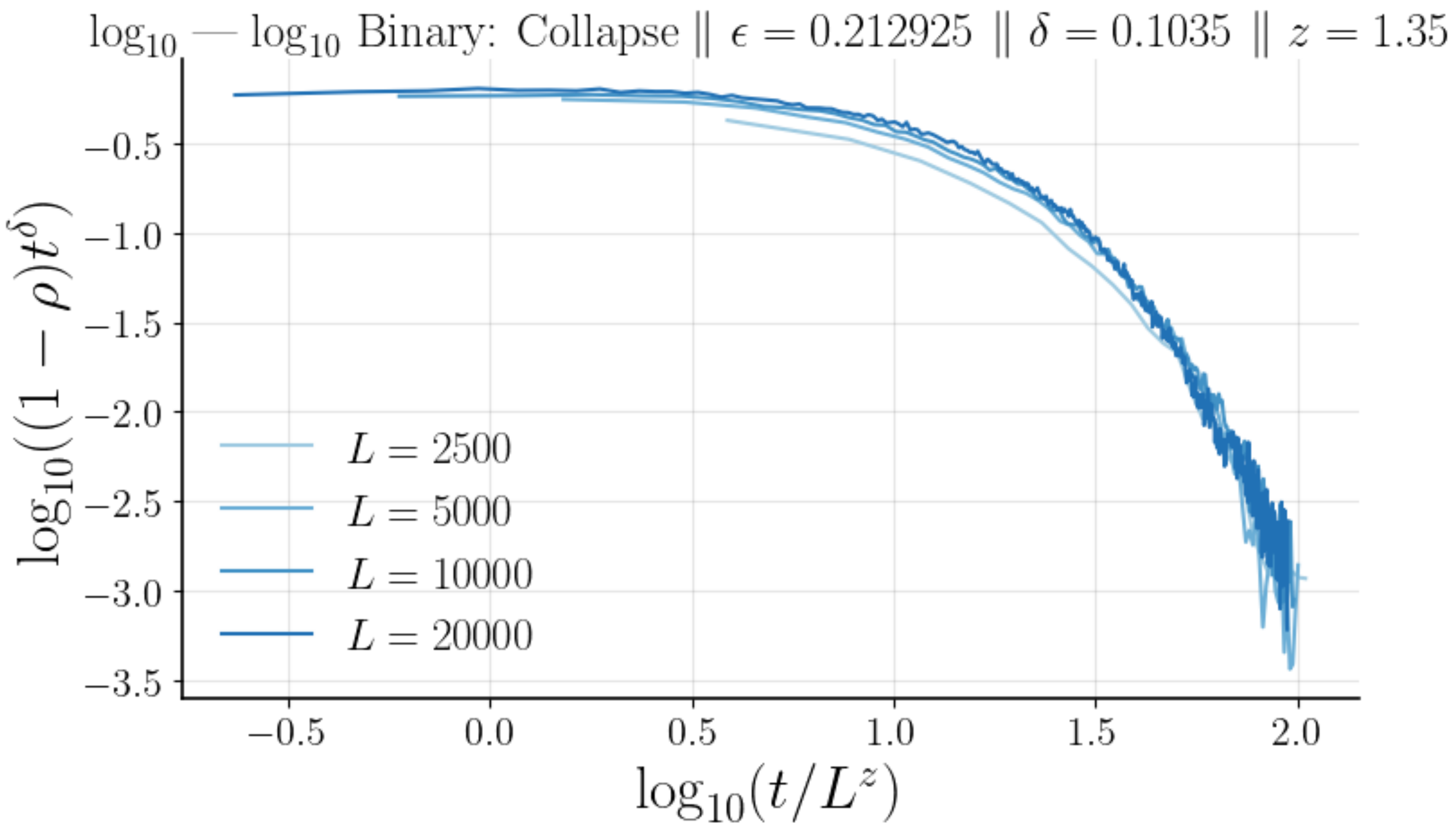
$\boxed{\nu_{\|}} \approx 2.3 \pm 0.2$ $\boxed{\beta} \approx 0.22 \pm 0.2$



The Binary Distribution

$\boxed{\delta} \approx 0.1035 \pm 0.002$ $\boxed{z} \approx 1.35$





Summary

Spin Chain:

$\boxed{a_c} \approx 0.7575 \pm 0.0005$

$\boxed{\nu_{||}} \approx 2.025 \pm 0.005$

$\boxed{\beta} \approx 0.23 \pm 0.01$

$\boxed{z} \approx 1.2 \pm 0.1$

Broad:

$\boxed{a_c} \approx 0.252375$

$\boxed{\nu_{||}} \approx 2.15 \pm 0.05$

$\boxed{\beta} \approx 0.25 \pm 0.02$

$\boxed{z} \approx 1.4$

Binary:

$\boxed{a_c} \approx 0.212925$

$\boxed{\nu_{||}} \approx 2.3 \pm 0.2$

$\boxed{\beta} \approx 0.22 \pm 0.2$

$\boxed{z} \approx 1.35$

Values for DP are: $\nu \approx 1.097, \;\; z \approx 1.58, \;\; \beta \approx 0.28$

Takeaways and Next Steps

New Universality Class? Looks like it but inconclusive for now.

> The close agreement on β between both the models is a very good sign.

> The near agreement on $\nu_{||}$ is further evidence.

> The disagreement on z between the models can be explained in a number of ways.

- The randomness could be too weak and we haven't properly flowed to the new UC (assuming it exists).

- More likely we just failed to identify the critical point of the spin chain because the finite sized effects occur too early (note the spin chain had L = 2,000, and Stavakay's model used L = 20,000!).

> The effect of the perturbations on the models was of a similar kind, suggesting a flow to a new universality class.

Takeaways and Next Steps

Next steps:

> Perform a proper analysis on the toy model such that we can get more meaningful error bars.

> Verify our results for the spin chain by preforming FSS on scalable observable: $\xi_\tau, \quad \lambda, \quad S_{\text{diff}}$

> Improve our estimate for a_c in the spin chain. We have thought of a way to get around our finite size restrictions.

Supplemental Next:

Thank you

> Professor Jedediah Pixley - Mentor

> Sarang Gopalakrishnan - Collaborator

> Manas Kulkarni - Collaborator

> David A. Huse - Senior Advisor

Questions?

Deriving the Equation of Motion from the Hamiltonian

$$\boxed{H} = \sum_{i=1}^{L} [J_x S_i^x S_{i+1}^x + J_y S_i^y S_{i+1}^y + J_z S_i^z S_{i+1}^z] \quad \dot{S}_n^\alpha = \{S_n^\alpha, H\}$$

$$\dot{P} = \{P, H\} := \partial_{q_i} P \partial_{p_i} H - \partial_{p_i} P \partial_{q_i} H$$

$$\{S_n^\alpha, S_m^\beta\} = \sum_{i=1}^{L} \sum_{q \in \{x,y,z\}} \partial_{r_i^q} S_n^\alpha \partial_{p_i^q} S_m^\beta - \partial_{p_i^q} S_n^\alpha \partial_{r_i^q} S_m^\beta$$

$$\{S_n^\alpha, H\} = \{S_n^\alpha, \sum_{i=1}^{L} [J_x S_i^x S_{i+1}^x + J_y S_i^y S_{i+1}^y + J_z S_i^z S_{i+1}^z]\} = \sum_{i=1}^{L} [J_x \{S_n^\alpha, S_j^x S_{j+1}^x\} + J_y \{S_n^\alpha, S_j^y S_{j+1}^y\} + J_z \{S_n^\alpha, S_j^z S_{j+1}^z\}]$$

$$\{S_n^\alpha, S_m^\beta S_l^\beta\} = \{S_n^\alpha, S_m^\beta\} S_l^\beta + S_m^\beta \{S_n^\alpha, S_l^\beta\}$$

$$\{S_n^\alpha, S_m^\beta\} = 0, \text{ if } n \neq m$$

$$\{S_n^\alpha, S_n^\beta\} = 0, \text{ if } \alpha = \beta$$

$$\boxed{\{S_n^\alpha, S_m^\beta\}, \ \alpha, \beta \in \{x, y, z\}(\{1, 2, 3\})}$$

$$\mathbf{S}_i = \mathbf{r}_i \times \mathbf{p}_i = (r_i^y p_i^z - r_i^z p_i^y, r_i^z p_i^x - r_i^x p_i^z, r_i^x p_i^y - r_i^y p_i^x)$$

$$\boxed{\dot{S}_n^\alpha = J_x[\{S_n^\alpha, S_n^x\}(S_{n+1}^x + S_{n-1}^x)] + J_y[\{S_n^\alpha, S_n^y\}(S_{n+1}^y + S_{n-1}^y)] + J_z[\{S_n^\alpha, S_n^z\}(S_{n+1}^z + S_{n-1}^z)]}$$

Deriving the Equation of Motion from the Hamiltonian

$$\{S_n^\alpha, S_m^\beta\} = 0, \text{ if } n \neq m$$

$$\{S_n^\alpha, S_n^\beta\} = 0, \text{ if } \alpha = \beta$$

$$\{S_n^\alpha, S_m^\beta\} = \sum_{i=1}^{L} \sum_{q\in\{x,y,z\}} \partial_{r_i^q} S_n^\alpha \partial_{p_i^q} S_m^\beta - \partial_{p_i^q} S_n^\alpha \partial_{r_i^q} S_m^\beta$$

$$\dot{S}_n^\alpha = J_x[\{S_n^\alpha, S_n^x\}(S_{n+1}^x + S_{n-1}^x)] + J_y[\{S_n^\alpha, S_n^y\}(S_{n+1}^y + S_{n-1}^y)] + J_z[\{S_n^\alpha, S_n^z\}(S_{n+1}^z + S_{n-1}^z)]$$

$$\dot{S}_n^x = J_y[\{S_n^x, S_n^y\}(S_{n+1}^y + S_{n-1}^y)] + J_z[\{S_n^x, S_n^z\}(S_{n+1}^z + S_{n-1}^z)]$$

$$\mathbf{S}_i = \mathbf{r}_i \times \mathbf{p}_i = (r_i^y p_i^z - r_i^z p_i^y, r_i^z p_i^x - r_i^x p_i^z, r_i^x p_i^y - r_i^y p_i^x)$$

$$\dot{S}_n^y = J_x[\{S_n^y, S_n^x\}(S_{n+1}^x + S_{n-1}^x)] + J_z[\{S_n^y, S_n^z\}(S_{n+1}^z + S_{n-1}^z)]$$

$$\{S_n^x, S_n^y\} = p_n^y r_n^x - p_n^x r_n^y = S_n^z \quad \{S_n^x, S_n^z\} = p_n^z r_n^x - p_n^x r_n^z = -S_n^y \quad \{S_n^y, S_n^z\} = S_n^x$$

The cyclic pattern continues and we have $\{S_n^\alpha, S_n^\beta\} = \epsilon_{\alpha\beta\gamma} S_n^\gamma$

$$\dot{S}_n^x = S_n^z[J_y(S_{n+1}^y + S_{n-1}^y)] - S_n^y[J_z(S_{n+1}^z + S_{n-1}^z)]$$

$$\dot{S}_n^y = -S_n^z[J_x(S_{n+1}^x + S_{n-1}^x)] + S_n^x[J_z(S_{n+1}^z + S_{n-1}^z)]$$

$$\dot{S}_n^z = S_n^y[J_x(S_{n+1}^x + S_{n-1}^x)] - S_n^x[J_y(S_{n+1}^y + S_{n-1}^y)]$$

$$\boxed{\frac{d\mathbf{S}_j}{dt} = [\mathbf{J} \circ (\mathbf{S}_{j-1} + \mathbf{S}_{j+1})] \times \mathbf{S}_j}$$

Deriving the Equation of Motion from the Hamiltonian

$$\frac{d\mathbf{S}_j}{dt} = [\mathbf{J} \circ (\mathbf{S}_{j-1} + \mathbf{S}_{j+1})] \times \mathbf{S}_j$$

$$f(x) := \sqrt{1 - x^2}$$

If we choose the canonical basis $s_i := S_i^x,\ S_i^y = f(s_i)\cos(\phi_i),\ S_i^z = f(s_i)\sin(\phi_i)$ and have $J_y = J_z = J$

This becomes

$$\begin{aligned}\dot{s}_i = f(s_i)J[&f(s_{i-1})\sin(\phi_i - \phi_{i-1})\\ &+f(s_{i+1})\sin(\phi_i - \phi_{i+1})]\end{aligned}$$

$$\begin{aligned}\dot{\phi}_i = -Jf'(s_i)[&f(s_{i+1})\cos(\phi_i - \phi_{i+1})\\ &+ f(s_{i-1})\cos(\phi_i - \phi_{i-1})]\\ &+ J_x(s_{i+1} + s_{i-1})\end{aligned}$$

Indeed ϕ_i is the conjugate momentum of s_i

Linear Analysis about Unstable Fixed Point

$$\mathbf{S}_j^0 = (0, \cos(2\pi j/N), \sin(2\pi j/N)) \qquad s_j^0 = 0,\ \phi_j^0 = 2\pi j/N,\ f(s_i^0) = 1$$

$$s_j = \delta s_j,\ \phi_j = 2\pi j + \delta\phi_j$$

$$\dot{\delta s_j} = \dot{s_j} \approx J\cos(2\pi/N)[2\delta\phi_j - \delta\phi_{j+1} - \delta\phi_{j-1}]$$

$$\dot{\delta\phi_j} = \dot{\phi_j} \approx -2J\delta s_j 2\cos(2\pi/N) + J_x[\delta s_{j+1} + \delta s_{j-1}]$$

$$\delta s_j = \frac{1}{L}\sum_k \delta s_k e^{-ikj}$$

$$\delta\phi_j = \frac{1}{L}\sum_k \delta\phi_k e^{-ikj}$$

$$,\ k = \frac{2\pi n}{L},\ n = 0, 1, ..., L-1$$

$$\dot{\delta s_k} = 2J\cos(2\pi/N)[1 - \cos(k)]\delta\phi_k \qquad B(k,N) = 2J\cos(2\pi/N)[1 - \cos(k)]$$

$$\dot{\delta\phi_k} = (-2J\cos(2\pi/N) + 2J_x\cos k)\delta s_k \qquad A(k,N) = (-2J\cos(2\pi/N) + 2J_x\cos k)$$

$$\begin{pmatrix}\dot{\delta s_k}\\ \dot{\delta\phi_k}\end{pmatrix} = \begin{pmatrix}0 & B(k,N)\\ A(k,N) & 0\end{pmatrix}\begin{pmatrix}\delta s_k\\ \delta\phi_k\end{pmatrix}$$

$$\boxed{\begin{pmatrix}\ddot{\delta s_k}\\ \ddot{\delta\phi_k}\end{pmatrix} = A(k,N)B(k,N)\begin{pmatrix}\delta s_k\\ \delta\phi_k\end{pmatrix}}$$

Linear Analysis about Unstable Fixed Point

$$\begin{pmatrix} \ddot{\delta s_k} \\ \ddot{\delta \phi_k} \end{pmatrix} = \lambda(k,N)^2 \begin{pmatrix} \delta s_k \\ \delta \phi_k \end{pmatrix} \qquad \lambda(k,N)^2 = -4J\cos(2\pi/N)(J\cos(2\pi/N) - J_x\cos(k))(1-\cos k)$$

If λ is positive, we have an exponential solutions $\delta s_k(t) = e^{\lambda(k,N)t}\delta s_k(0),\ \ \delta\phi_k(t) = e^{\lambda(k,N)t}\delta\phi_k(0)$.

Inspection of λ shows that there exist stable modes for all N.

The Control Map and Phase Diagram

If we preform the control map every unit tick of time, then it makes sense to write

$$\hat{C}\mathbf{S}_j(t') = \frac{(1-a)\mathbf{S}_j^0 + a\mathbf{S}_j(t')}{|(1-a)\mathbf{S}_j^0 + a\mathbf{S}_j(t')|}$$

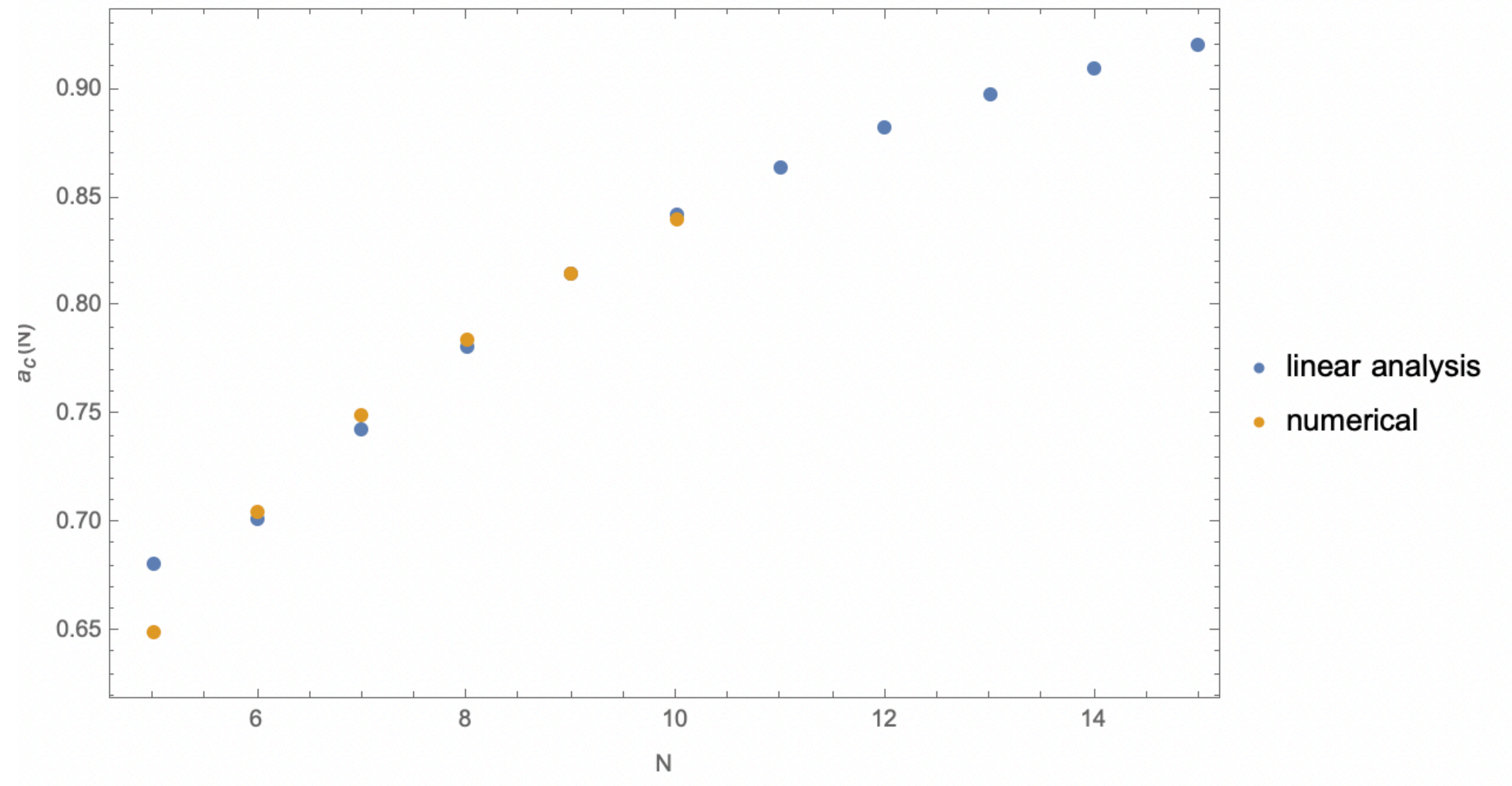
$$\delta s_k(t) = (e^{\lambda(k,N)})^t \delta s_k(0), \;\; \delta\phi_k(t) = (e^{\lambda(k,N)})^t \delta\phi_k(0)$$

To linear order $\hat{C}\delta s_k = a\delta s_k, \;\; \hat{C}\delta\phi_k = a\delta\phi_k$

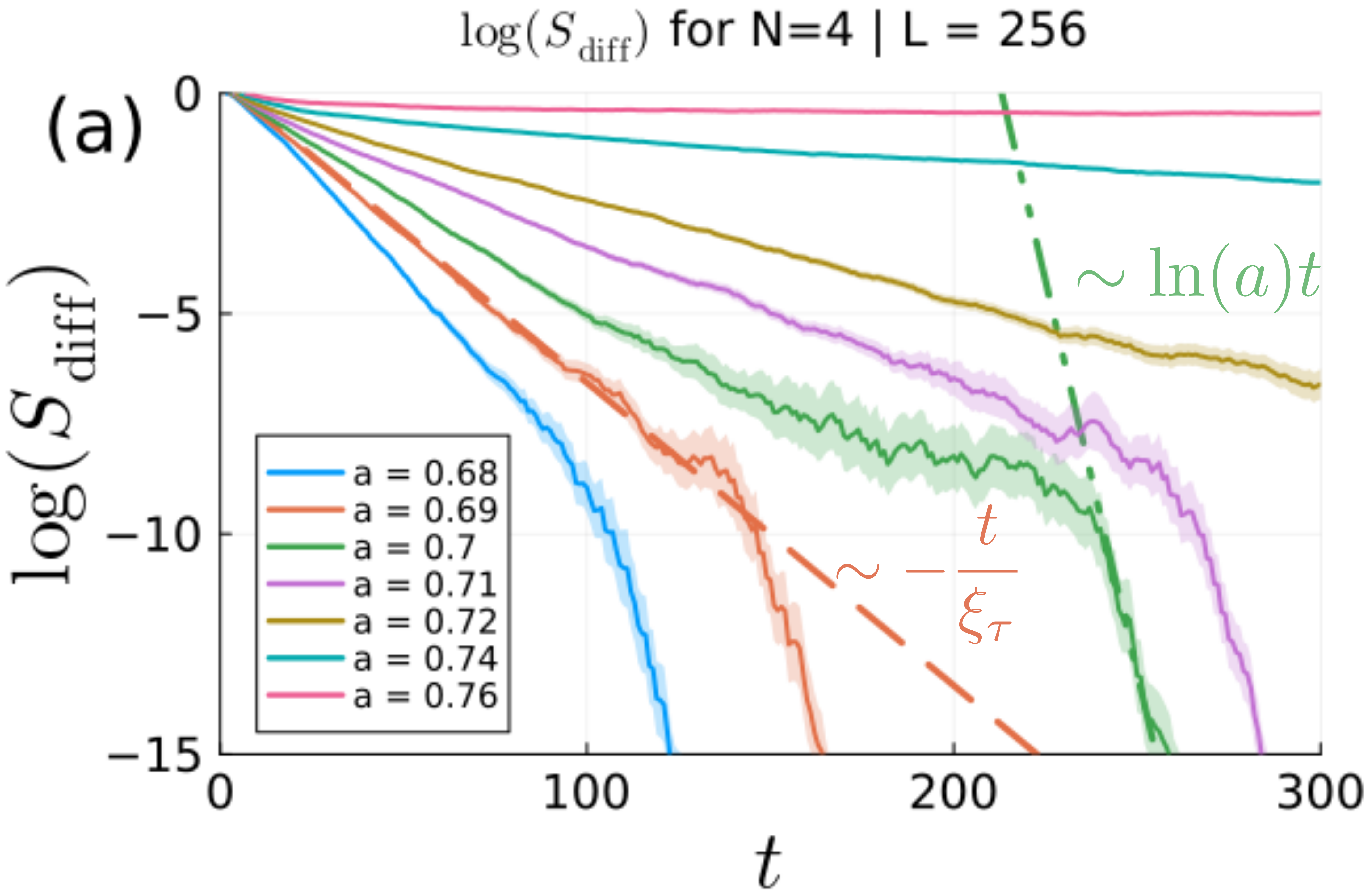
Thus the full evolution can be expressed as

$$\delta s_k(t) = (ae^{\lambda(k,N)})^t \delta s_k(0), \;\; \delta\phi_k(t) = (ae^{\lambda(k,N)})^t \delta\phi_k(0)$$

We can now naturally define the critical point to be $a_c, ae^{\max_k \lambda(k,N)} = 1$



Supplemental: Temporal Correlation Length



As a result of the evolution

$$S_{\text{diff}}(t) \sim e^{-t/\xi_\tau}$$

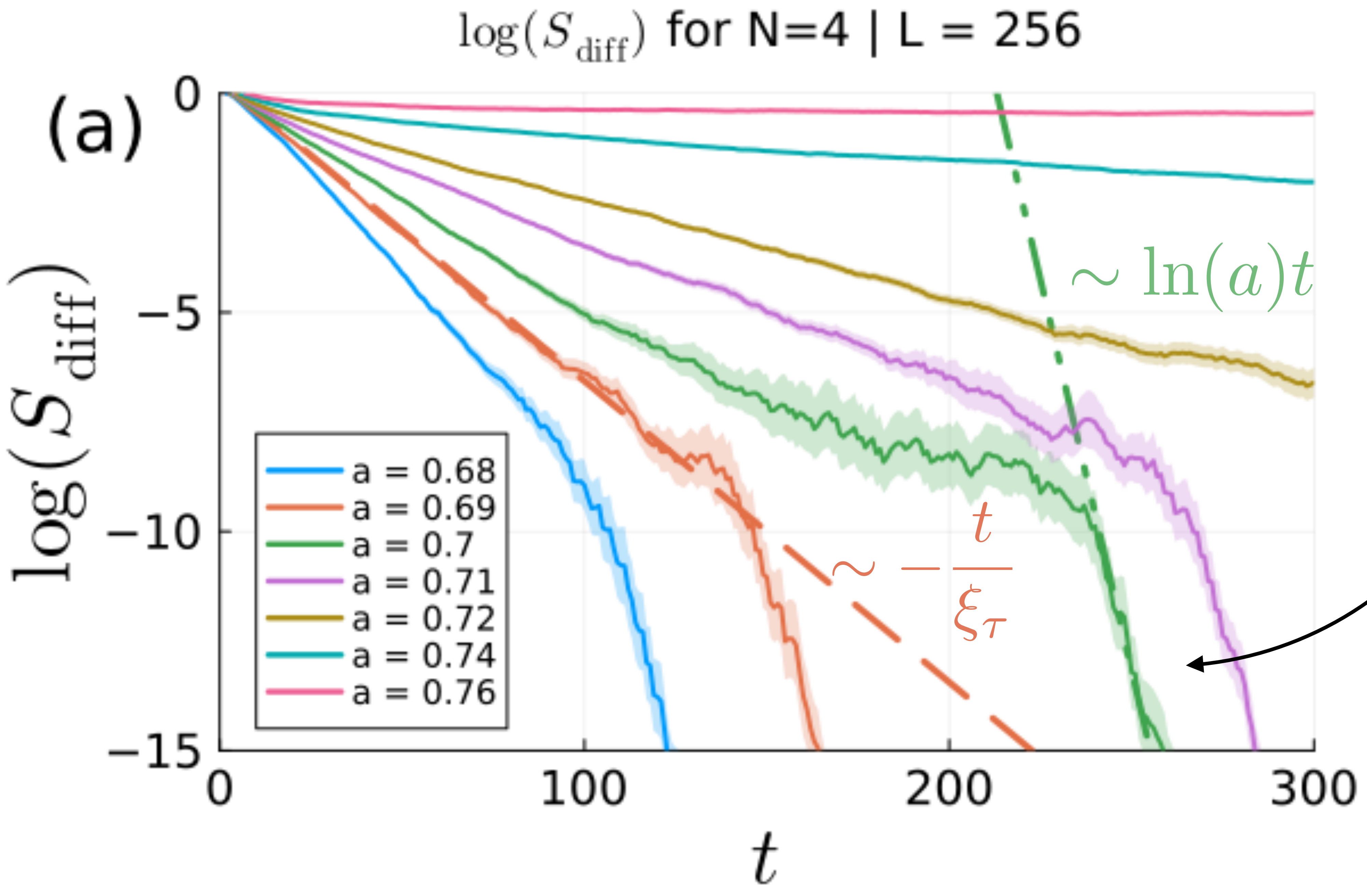
With
$$\begin{cases} 1/\xi_\tau > 0\ , & a << a_c \\ 1/\xi_\tau = 0\ , & a >> a_c \end{cases}$$

From theory (Goldenfeld Lectures)

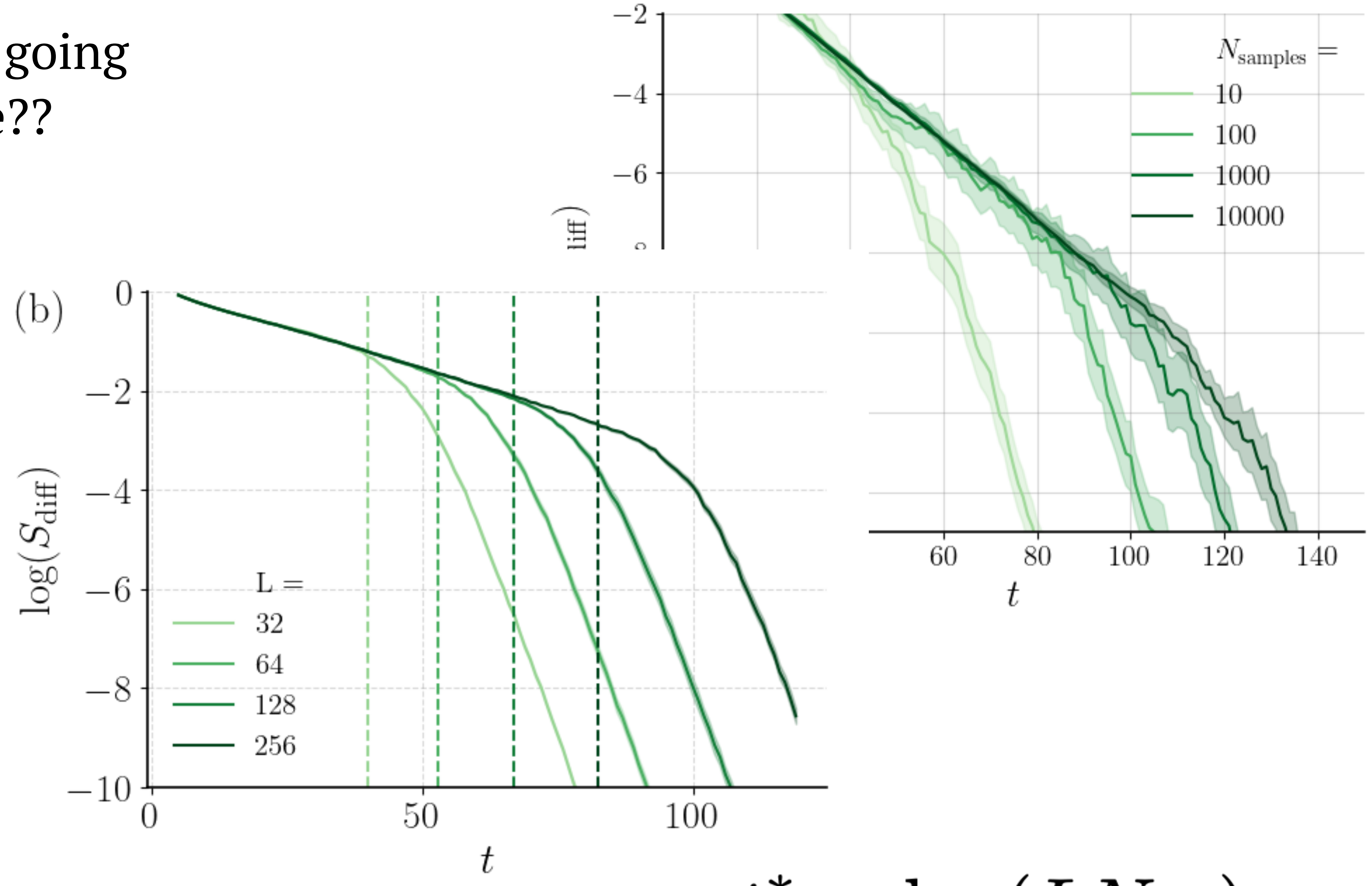
$$\xi_\tau \sim \xi^z \sim |a - a_c|^{-z\nu_\perp} \qquad a \approx a_c$$

We sought and found the expected $\quad 1/\xi_\tau \sim |a - a_c|^{\nu_\parallel} \qquad \nu_\parallel := z\nu_\perp$

Supplemental: The Shoulder



What's going on here??



It is a FS effect, and is also delayed by considering more initial conditions

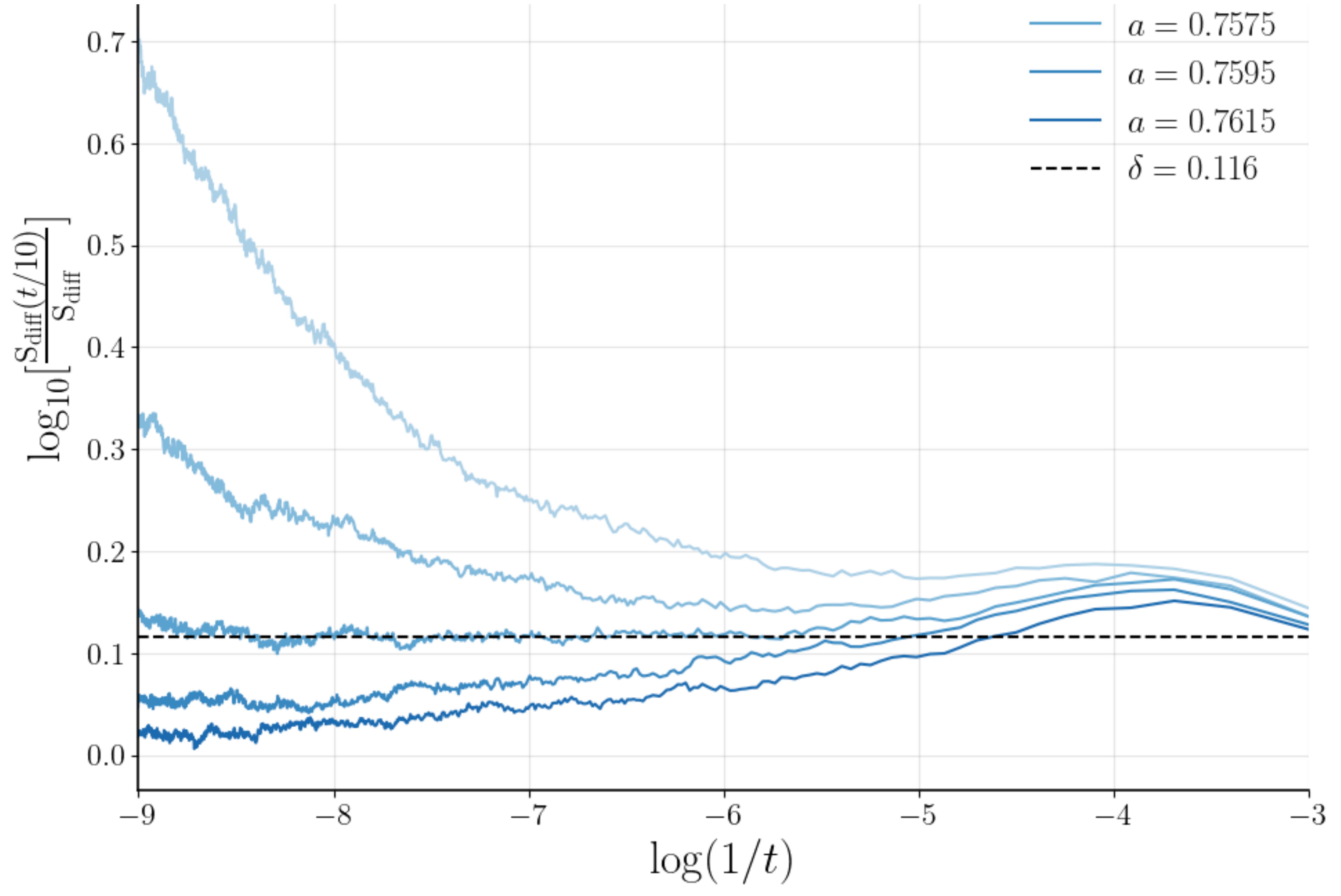
$$t^* \sim \log(L N_{\mathrm{IC}})$$

Supplemental: The $t/L^z << 1$ Regime: Determining $a_c, \ \nu_\parallel, \beta$

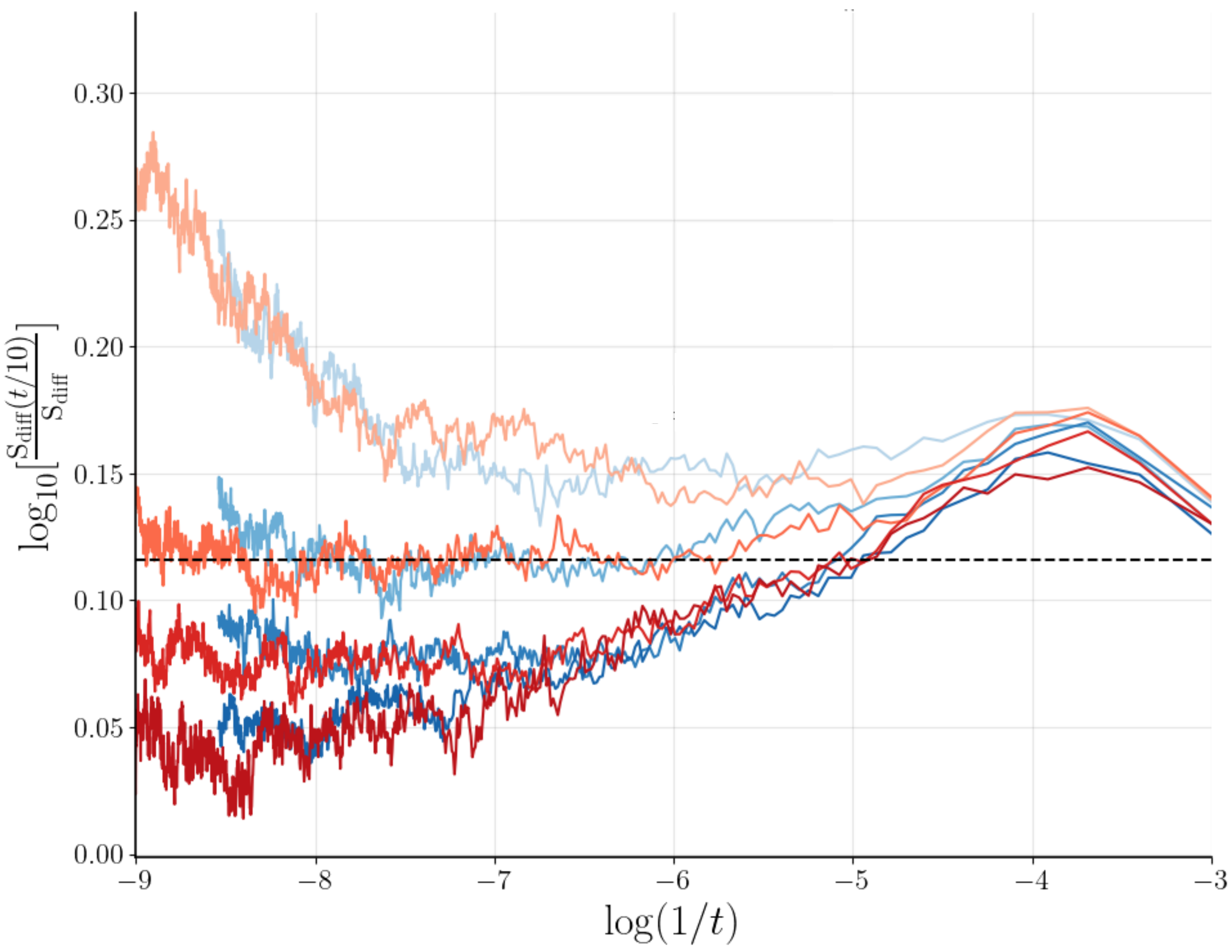
If we consider the case where $a = a_c \implies \Delta = 0$

$$\boxed{S_{\text{diff}}(t,\Delta) \sim t^{-\beta/\nu_\parallel}\Phi(\Delta t^{1/\nu_\parallel})}$$

$$\log_b\left(\frac{n(t/b)}{n(t)}\right) = \log_b(n(t/b))) - \log_b(n(t)) \sim \log_b(t^{-\delta}/b^{-\delta}) - \log_b(t^{-\delta}) = \delta$$



Determining The $t/L^z << 1$ Regime: Determining a_c, $\nu_{\|}, \beta$



$$S_{\text{diff}}(t,\Delta) \sim t^{-\beta/\nu_{\|}}\,\Phi(\Delta t^{1/\nu_{\|}})$$

$$0.7559 <= a <= 0.7599$$

The orange shows L = 2000

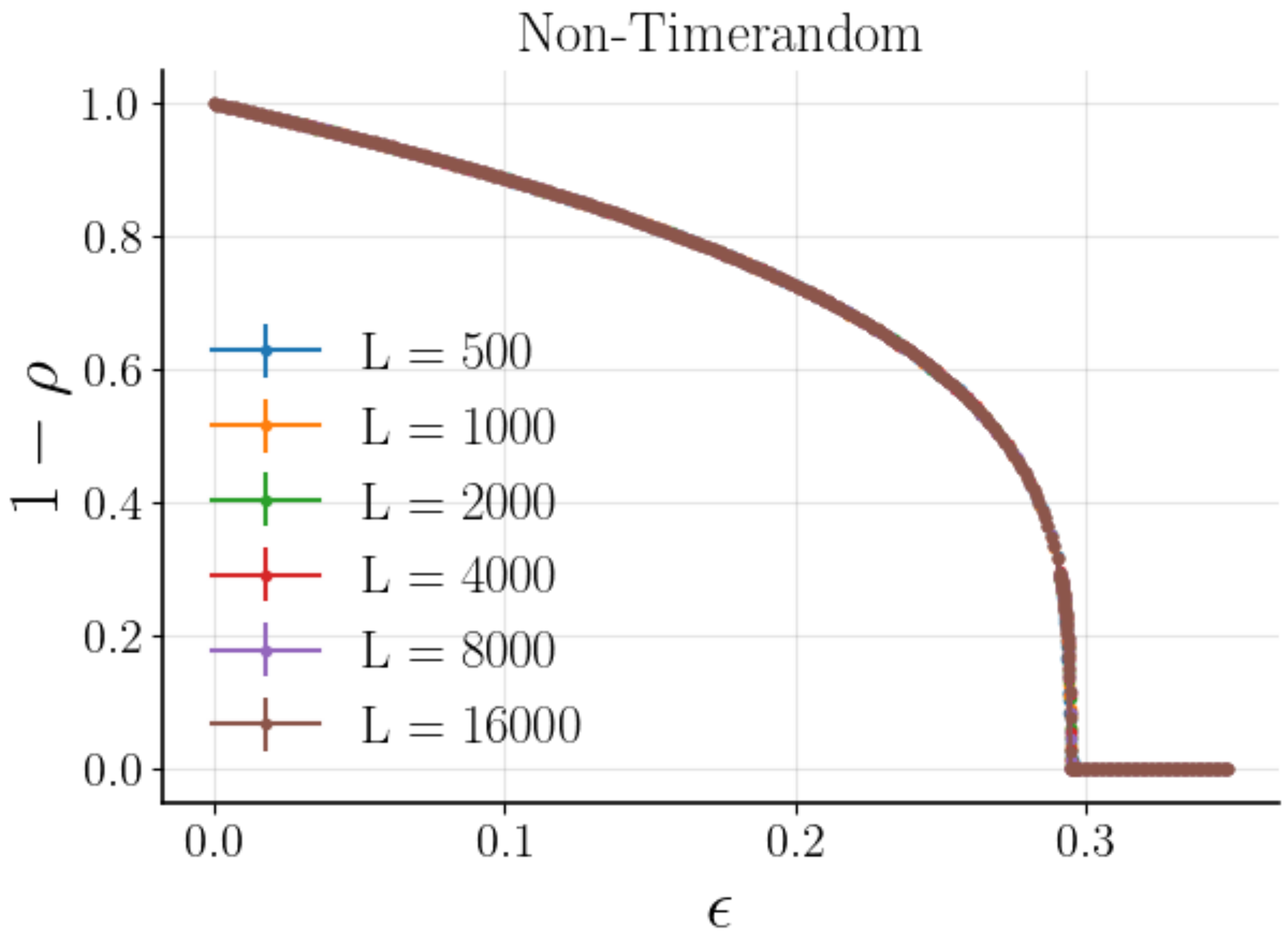
The blue shows L = 1024

Note the pealing that starts to occur around

$$t = e^{8}$$

This is where we estimate the finite sized effects begin to kick in and the power law scaling ends

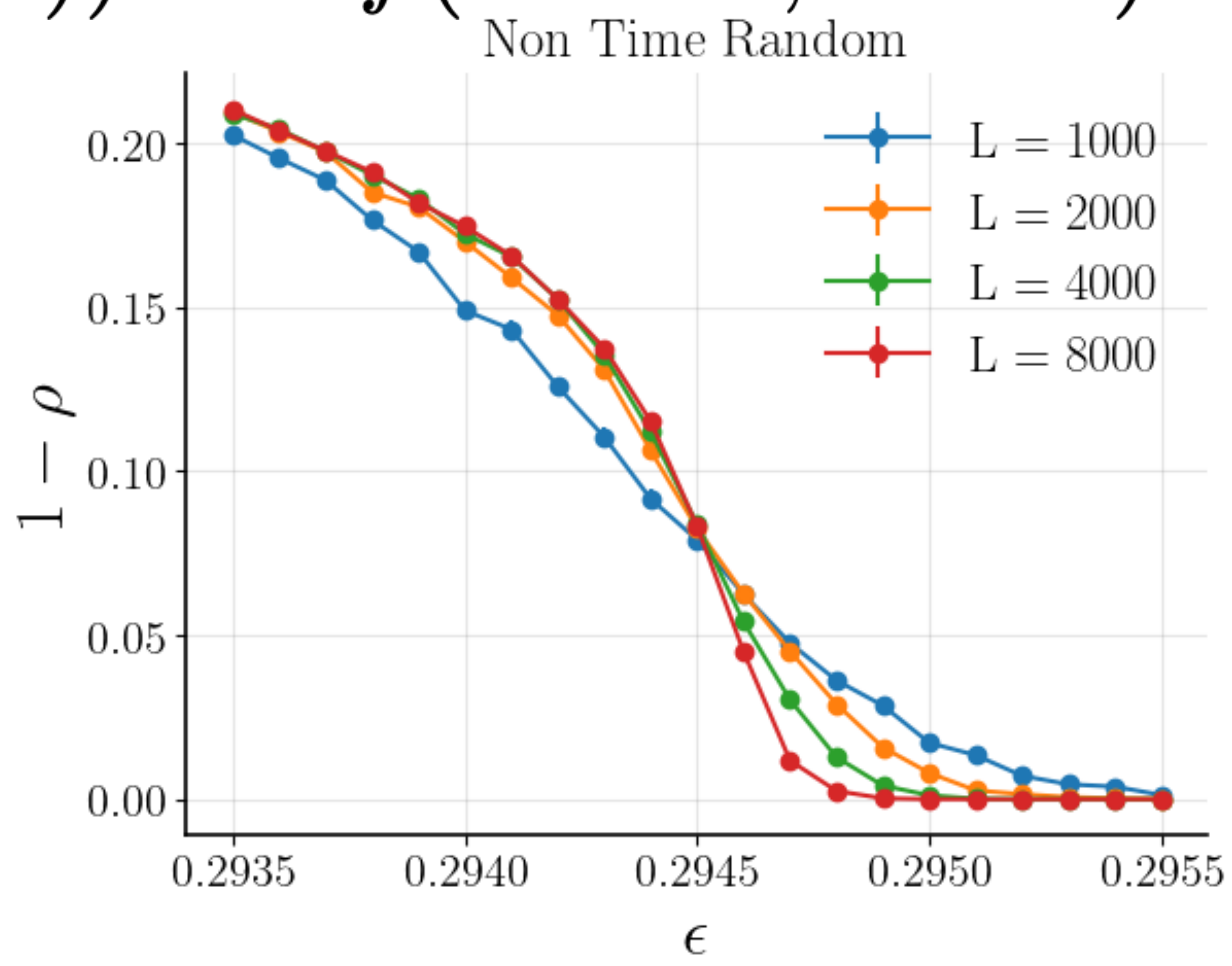
The Original Model



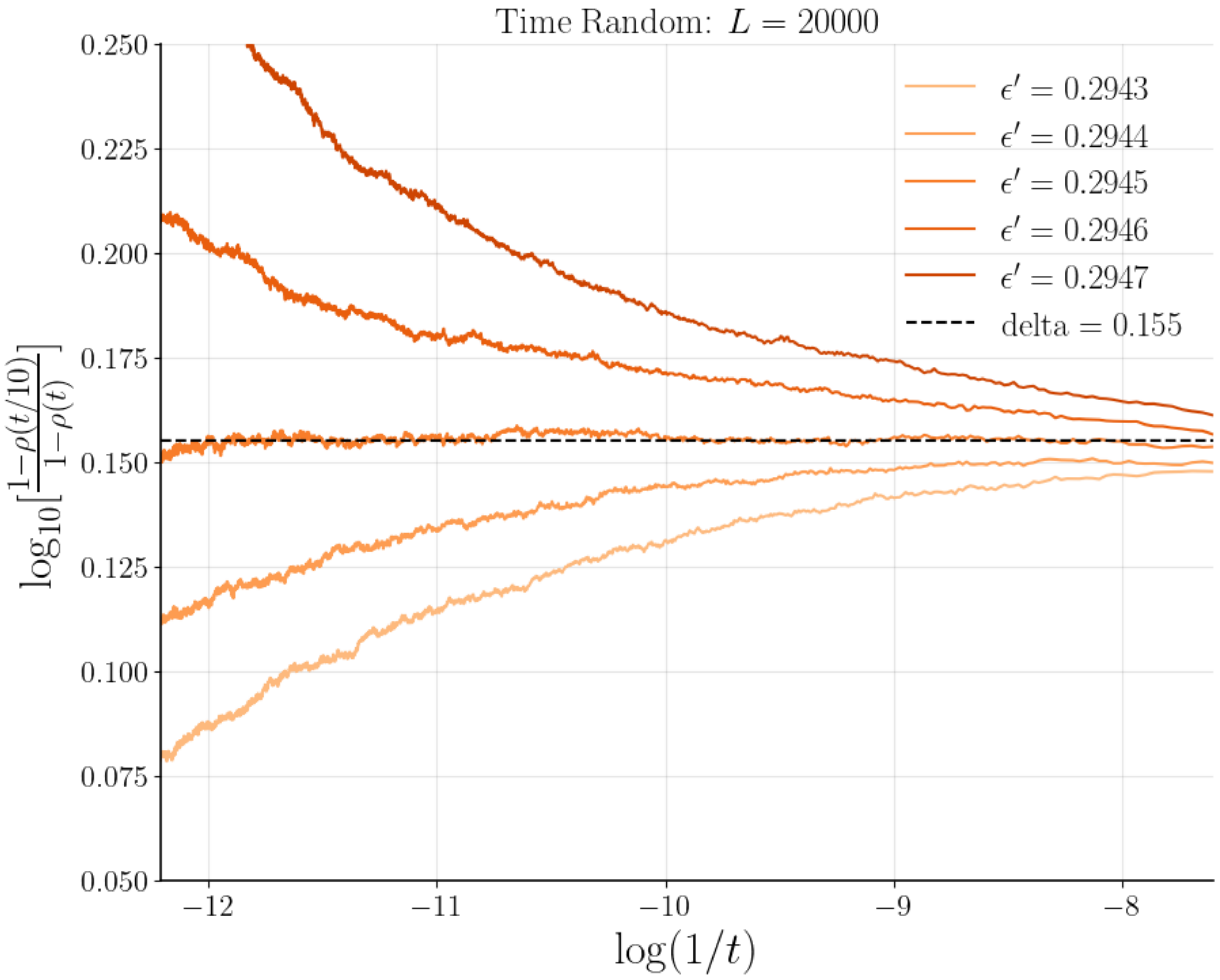
$$1-\rho(t;\Delta) \sim t^{-\beta/\nu_\parallel} f(\Delta t^{1/\nu_\parallel}, t^{\nu_\perp/\nu_\parallel})$$

$$\delta := \beta/\nu_\parallel$$

$$(1-\rho(t;\Delta))t^{\delta} \sim f(\Delta t^{1/\nu_\parallel}, t^{\nu_\perp/\nu_\parallel})$$

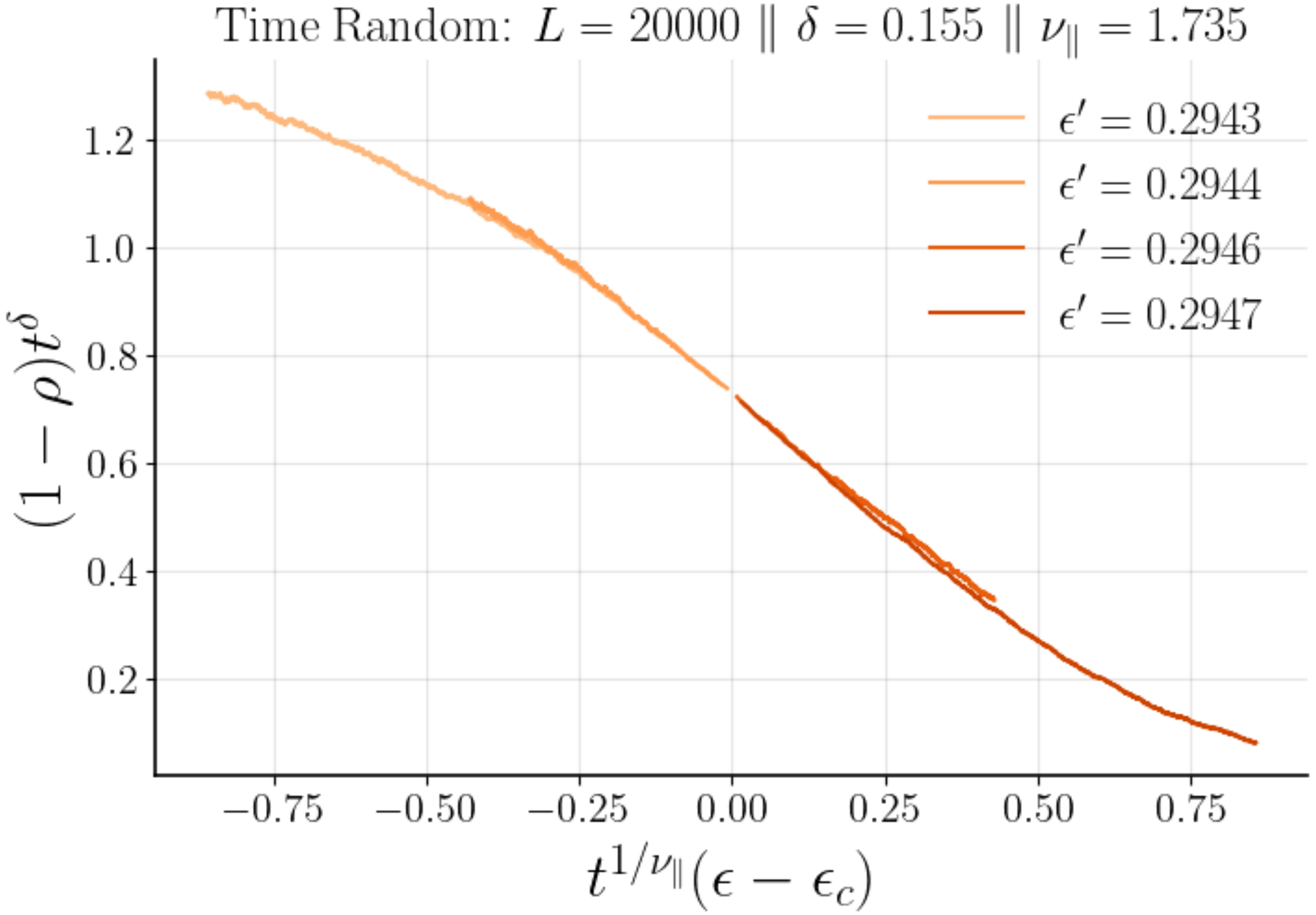


The Original Model (Deterministic)



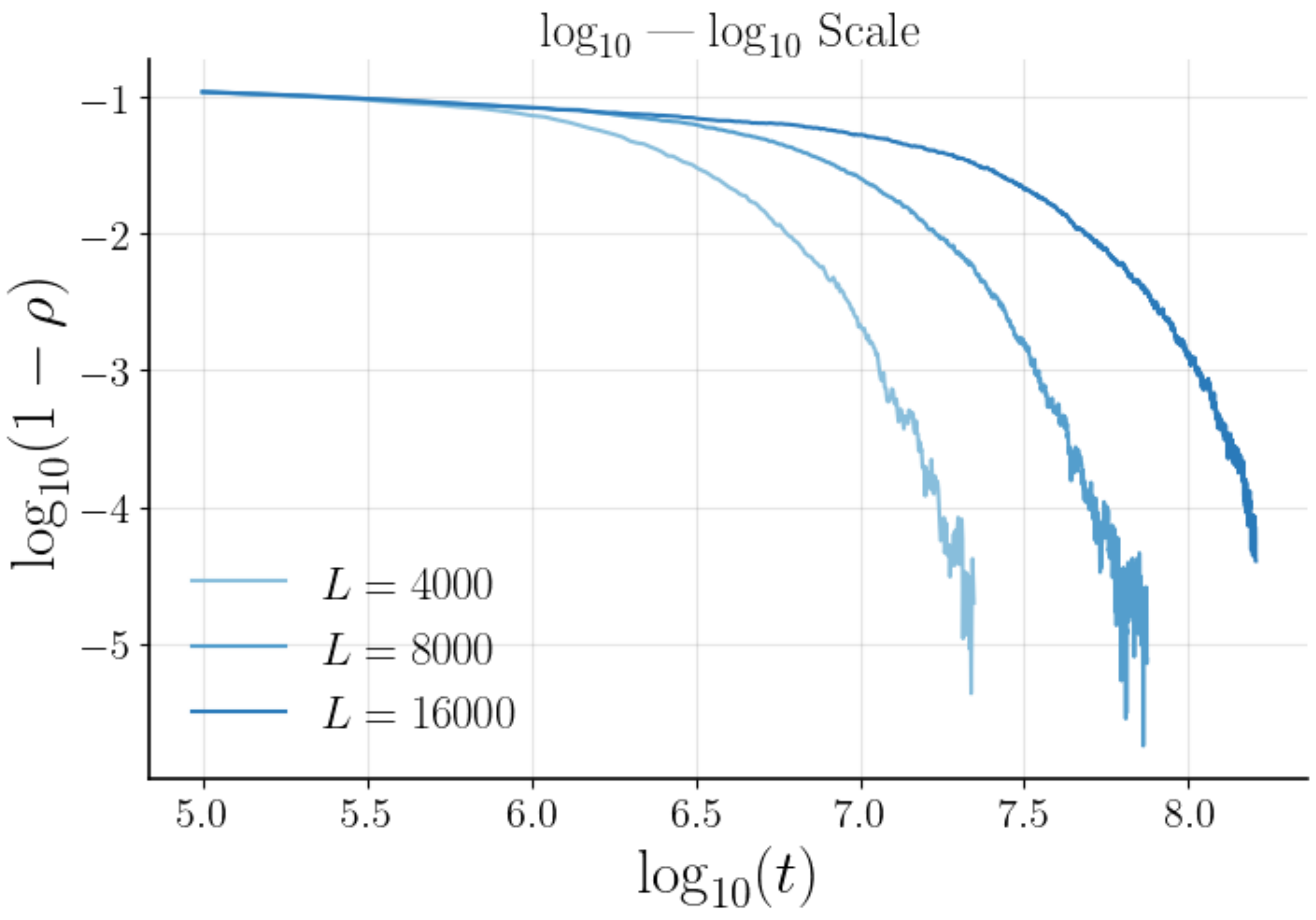
$\boxed{a_c} = 0.2945$

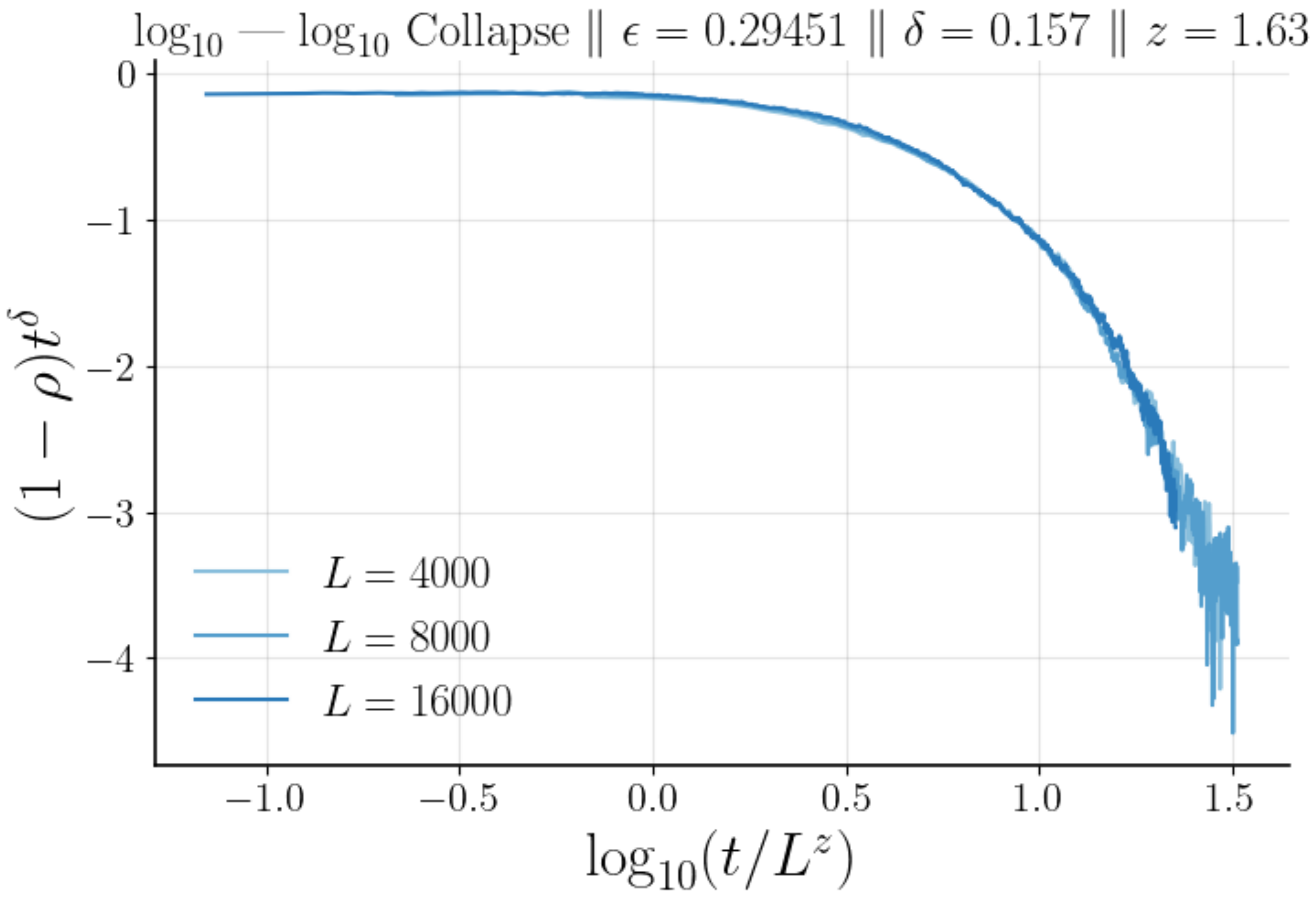
$\boxed{\delta} = 0.155 \quad \boxed{\nu_\parallel} = 1.735 \implies \boxed{\beta} = 2.69$



The Original Model (Deterministic)

$\boxed{\delta} = 0.157 \quad \boxed{z} = 1.63$





FSS to find verify exponents - ν, z, and β

We will approximate the critical exponents by preforming finite sized scaling (FSS) on three observables:

Temporal correlation length

$$\xi_\tau \sim L^z f[(a - a_c)L^{1/\nu}]$$

$$\longrightarrow \boxed{a_c}, \boxed{\nu}, \boxed{z}$$

$$\xi_\tau \sim |a - a_c|^{-z\nu}$$

The order parameter

$$S_{\text{diff}}(t = L^z, a) := \langle|\delta \mathbf{S}_j|\rangle \sim L^{\nu/\beta} g[(a - a_c)L^{1/\nu}]$$

$$\longrightarrow \boxed{a_c}, \boxed{\nu}, \boxed{\beta}\ \boxed{z}$$

The STD of the Lyapunov exponent

$$\mathrm{Std}(\lambda(t = L^z, a)) \sim h[(a - a_c)L^{1/\nu}]$$

$$\longrightarrow \boxed{a_c}, \boxed{\nu}, \boxed{z}$$

$$D(x,t) := 1 - \langle \mathbf{S}_x^A \cdot \mathbf{S}_x^B \rangle \sim e^{\lambda_a t}$$

Values for DP are: $\nu \approx 1.097$, $z \approx 1.58$, $\beta \approx 0.28$

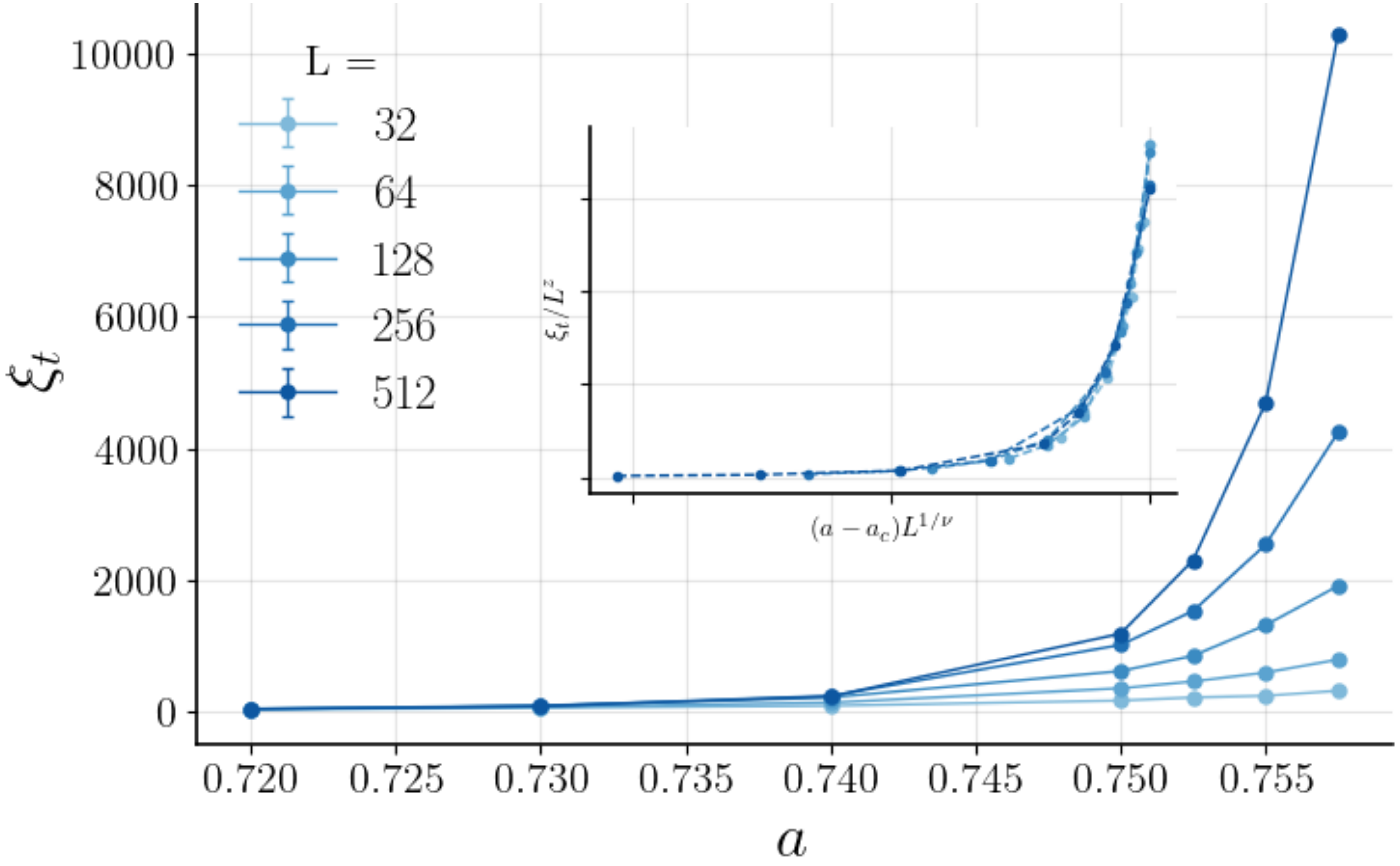
Temporal Correlation Length Before the Transition: FSS

$$\xi_\tau \sim L^{\boxed{z}} f\left[(a - \boxed{a_c})L^{1/\boxed{\nu}}\right]$$

$\boxed{\nu} := \nu_\parallel / z \; (= \nu_\perp) = 1.6$

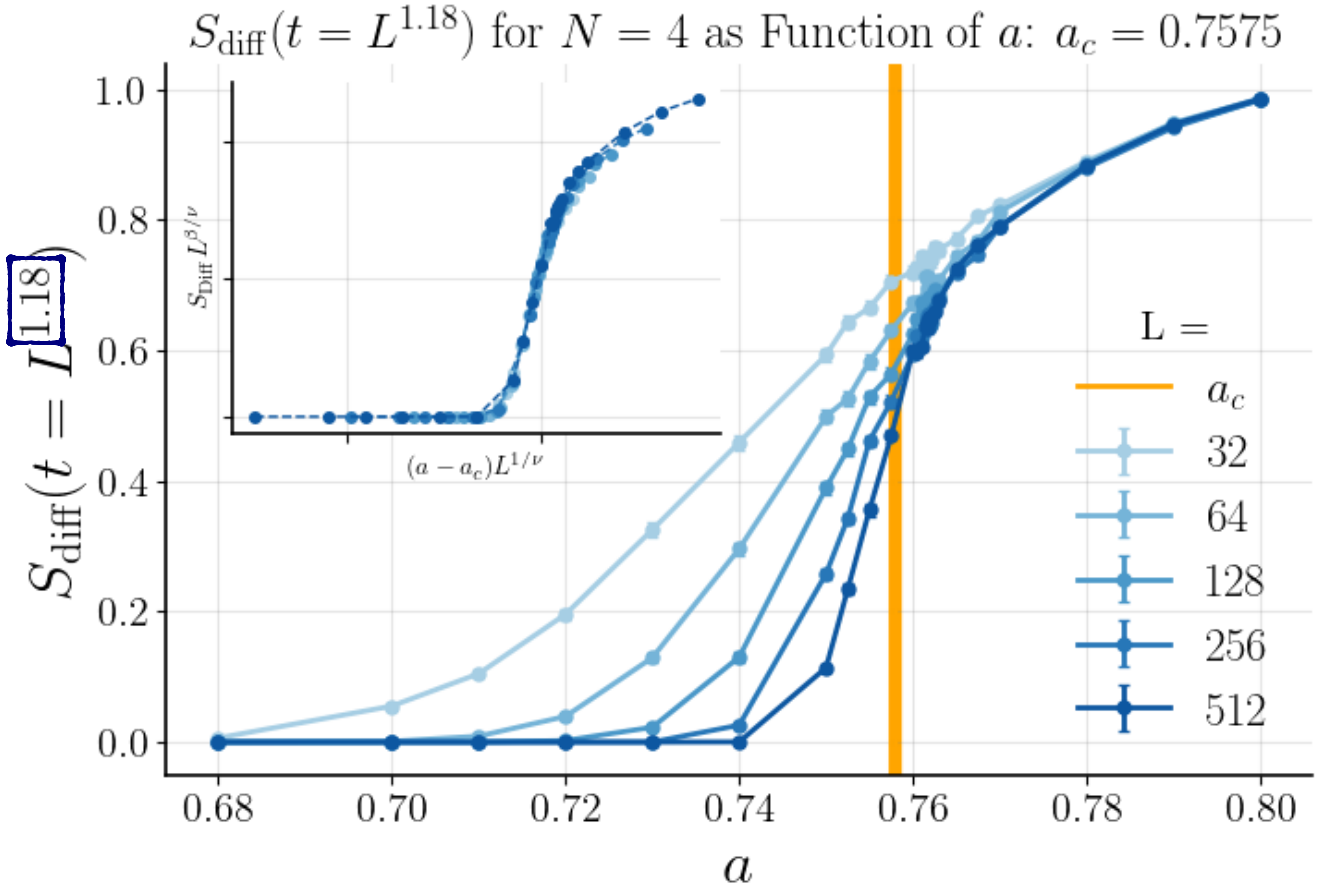
$\boxed{a_c} = 0.7575$

$\boxed{z} \approx 1.3 \pm 0.1$



The Order Parameter Across the Transition: FSS

$$S_{\rm diff}(t = L^{\boxed{z}}, a) := \langle|\delta \mathbf{S}_j|\rangle \sim L^{\boxed{\nu}/\boxed{\beta}} g[(a - \boxed{a_c})L^{1/\boxed{\nu}}]$$

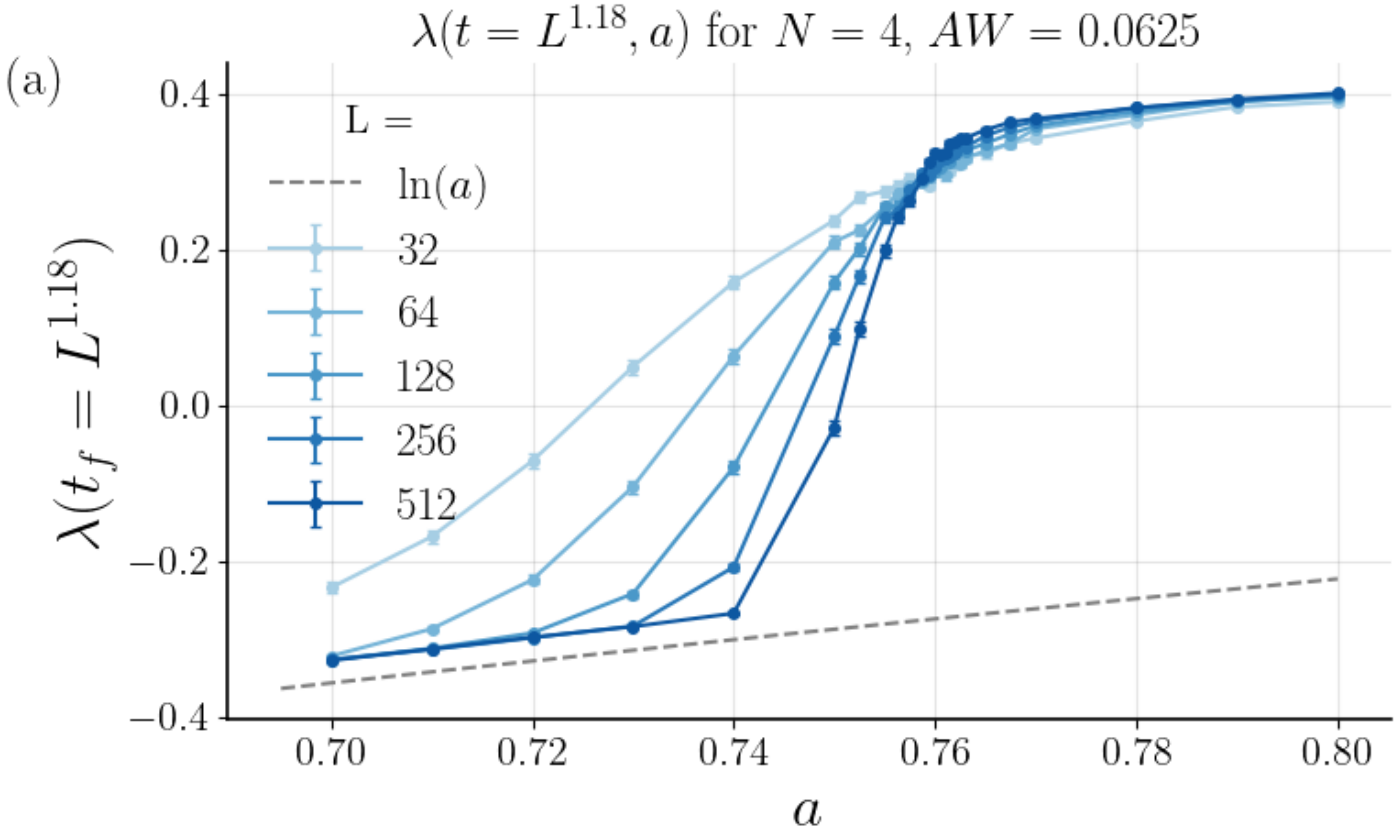


$\boxed{a_c} = 0.7575$

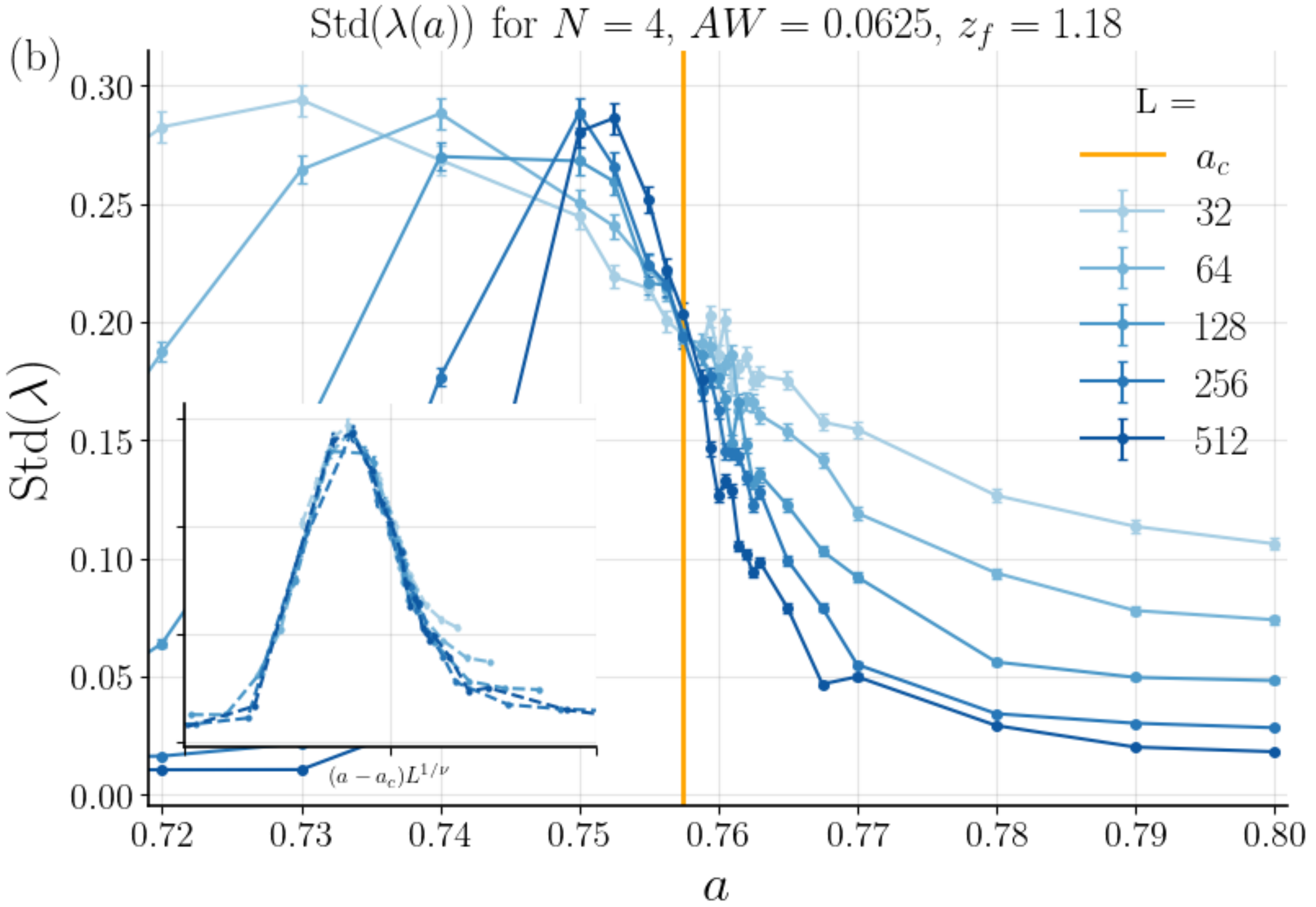
$\boxed{z} = 1.18$

$\boxed{\nu} = 1.7$

The Standard Deviation of Lyoponov Across the Transition



$$\mathrm{Std}(\lambda(t = L^{\boxed{z}}, a)) \sim h[(a - \boxed{a_c})L^{1/\boxed{\nu}}]$$



$\boxed{a_c} = 0.7575$

$\boxed{z} = 1.18$

$\boxed{\nu} = 1.7$

Summary of FSS

Temporal correlation length

$$\xi_\tau \sim L^z f[(a-a_c)L^{1/\nu}]$$

$$\boxed{\nu} := \nu_{\|}/z \ (= \nu_\perp) \ = 1.6$$

$$\boxed{a_c} = 0.7575$$

$$\boxed{z} \approx 1.3 \pm 0.1$$

The order parameter

$$S_{\mathrm{diff}}(t = L^z, a) := \langle |\delta \mathbf{S}_j| \rangle$$
$$\sim L^{\nu/\beta} g[(a-a_c)L^{1/\nu}]$$

$$\boxed{a_c} = 0.7575$$

$$\boxed{z} = 1.18$$

$$\boxed{\nu} = 1.7$$

The STD of the Lyapunov exponent

$$\mathrm{Std}(\lambda(t = L^z, a))$$
$$\sim h[(a-a_c)L^{1/\nu}]$$

$$\boxed{a_c} = 0.7575$$

$$\boxed{z} = 1.18$$

$$\boxed{\nu} = 1.7$$

Values for DP are: $\nu \approx 1.097, \ z \approx 1.58, \ \beta \approx 0.28$

Thank You

Questions?